Component Decomposition Analysis for Hyperspectral Anomaly Detection

Shuhan Chen, *Member, IEEE*, Chein-I Chang, *Life Fellow, IEEE*, and Xiaorun Li

***Abstract*— Low-rank and sparse representation (LRaSR)-based approaches have been widely used for anomaly detection (AD). Their central ideas are to minimize the rank of the low-rank space constrained to predetermined values, while using various regularization parameters to control the sparse representation. Three key issues arise from LRaSR. The first is how to determine the constrained rank. The second is an appropriate selection of regularization parameters. The third one is the detector used for AD. This article presents a new but rather simple competing model, called component decomposition analysis (CDA) which represents a data space X as a linear orthogonal decomposition of three components, $\mathbf{X} = \mathbf{PC}^m + \mathbf{IC}^j + \mathbf{N}$ with m principal components, $\mathbf{PC}^m$, generated by principal component analysis (PCA) and j independent components, $\mathbf{IC}^j$, generated by independent component analysis (ICA) plus a noise component N. CDA offers several advantages over LRaSR. First, CDA uses well-known component analysis techniques to decompose the dataset without solving constrained optimization problems. Second, the values of m and j can be automatically determined by virtual dimensionality (VD) and a minimax-singular value decomposition (MX-SVD). To better extract anomalies from the $\mathbf{IC}^j$ component space, the concept of sparsity cardinality (SC) is further incorporated into CDA to derive a CDASC anomaly detector (CDASC-AD). The experimental results demonstrate that CDASC-AD is very competitive against the LRaSR-based models and performs well in hyperspectral AD.**

***Index Terms*— Component decomposition analysis (CDA), component decomposition analysis sparsity cardinality-based anomaly detector (CDASC-AD), independent component analysis (ICA), low-rank and sparse representation (LRaSR), principal components analysis (PCA), sparsity cardinality (SC), virtual dimensionality (VD).**

I. Introduction

ANOMALY detection (AD) is a fundamental task in hyperspectral data exploitation which can uncover unknown subtle material substances that cannot be identified by prior knowledge or visual inspection [1] and [2, Ch. 5, 14–17]. It can be considered as a dual theory of hyperspectral target detection [3]. One most widely used approach is the one developed by Reed and Yu [4], referred to as Reed–Xiaoli anomaly detector (RX-AD) in this article, also known as Mahalanobis distance, which used the global sample auto-covariance matrix **K** to perform AD. It was later extended to an anomaly detector developed by Chang and Chiang [5], referred to as sample correlation matrix (**R**)-based anomaly detector (R-AD) in [2], which replaced **K** with the global sample auto-correlation matrix **R**. The details of the theoretical derivations of RX-AD and R-AD from a hyperspectral imaging perspective can be found in [3]. Since RX-AD and R-AD were introduced, many RX/R-AD-type detectors and variants have been developed and reported in the past. Examples include local RX-AD [6]–[7], support vector data description (SVDD) [8], weighted-RX-AD and linear filter-based RX-AD in [9], a sliding window [10], a dual-window-based eigen separation transform (DWEST) developed in [11] followed by collaborative representation-based anomaly detector (CRD) which also used dual windows in [12], and multiple-window AD (MWAD) in [13] by making use of multiple windows to perform AD adaptively, guided filtering-based anomaly detector [14], spectral–spatial feature extraction-based anomaly detector [15], and a nonlinear anomaly detector, kernel-based anomaly detector (KRX-AD) [16]. In addition, there were also anomaly detectors developed for various applications, such as unlabeled classification [17] and real-time processing of AD [18]. Most recently, Liu *et al.* [19] extended AD in the Gaussian background with an unknown covariance matrix from two equivalent adaptive generalized likelihood ratio test (GLRT)-based detectors. Interestingly, Chang [3] showed that GLRT can be indeed obtained without using Gaussian assumption. Yuan *et al.* [20] proposed an AD method by constructing a vertex- and edge-weighted graph to detect anomalies with better accurate detection performance and better robustness to noise where no background distributions are required. Li *et al.* [21] developed an attribute and edge-preserving filtering-based detection method (AED) by fusing the multiscale information extracted by attribute and edge-preserving filters. Tao *et al.* [22] considered a new approach, fractional Fourier entropy (FrFE)-based AD method including three steps: 1) FrFT for multiple domain feature extraction; 2) order selection by FrFE maximization; and 3) AD by RXD in a p-fractional domain. However, as pointed out in [2] and [3], a crucial difference between target detection and AD is that the former is an active target detection which assumes that the targets to be detected must

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be known, while the latter is a passive target detection which does not require any target knowledge at all. This taxonomy was initially used by military intelligence applications where active target detection corresponds to reconnaissance which requires the prior target knowledge to be used for target detection as opposed to passive target detection corresponding to surveillance without knowing any target knowledge. As a result, active target detection is generally evaluated by the detectability of known targets to be detected. That is why binary simple hypothesis testing is used to detect targets in noise. In contrast, since AD does not know the targets it detects, its performance must rely on its ability in background suppression (BS). By virtue of BS, the contrast enhancement between background (BKG) and anomalies can bring targets of interest out of BKG to light. So, two key features that should have significant impacts on AD are BS and detection enhancement of anomalies. These two issues have triggered the most recent trends of developing and designing the AD algorithms.

As for the BS issue, one of most recent efforts is low-rank representation (LRR)-based approaches which considered BKG to be represented by a low-rank space. A typical example is [23] which developed LRR methods for AD based on the assumption that a hyperspectral dataset can be effectively decomposed into a low-rank component for BKG and a sparse component for anomalies by imposing a spatial constraint on the LRR coefficients smoothed by single or multiple local window strategies. Two other approaches are also of interest. One is BKG estimation using BKG training samples such as deep brief networks [24]. The other is BKG reconstruction by autoencoder (AE) [25] and generative adversarial network (GAN) [26], [27]. In either case, the estimation/reconstruction error is assumed to be the main cause incurred by anomalies.

As for AD enhancement, Qu *et al.* [28] developed a Gaussian mixture model (GMM)-based approach to anomaly extraction and an effective GMM-based weighting approach for fusing the extracted anomaly maps which can be further rectified using a guided filter to obtain the final AD map.

The main focus of the above-mentioned approaches is centered on BKG estimation/reconstruction or AD enhancement. A more effective means is to jointly consider BKG and AD enhancement issues together. Chang *et al.* [29] developed a BKG–anomaly component projection and separation optimized filter (BASO) to separate anomalies from BKG. In recent studies, low-rank and sparse representation (LRaSR) model-based methods have been developed to separate BKG and anomalies into two different separate spaces. Several specific models of particular interest have been investigated for AD in the literature and briefly reviewed as follows.

Generally speaking, LRaSR formulates AD as the following basic optimization problem [30]:

$$\min_{\mathbf{S},\mathbf{E}}\left\{\text{rank}(\mathbf{S})+\lambda||\mathbf{E}||_{2,1}\right\} \text{ subject to } \mathbf{X}=\mathbf{DS}+\mathbf{E} \quad (1)$$

where $\mathbf{D}$ is the BKG dictionary with $\mathbf{S}$ being the representation coefficients and $\mathbf{E}$ is its residual, and $||\cdot||_{2,1}$ is the $l_{2,1}$-norm defined as the sum of l_2-norm of the column of a matrix. The parameter λ is used to regularize the sparse representation $\mathbf{E}$. In other words, $\mathbf{DS}$ and $\mathbf{E}$ represent the BKG and anomalies, respectively. Many works based on (1) have been reported.

As an alternative to LRaSR in (1), robust principal component analysis (RPCA) developed in [31] can be used as a blind source separation technique to decompose the data matrix $\mathbf{X}$ into

$$\mathbf{X}=\mathbf{L}+\mathbf{S} \quad (2)$$

in matrix decomposition. An issue arising from the model in (2) is that it does not separate noise from sparse space. Most interestingly, a more attractive model, called low-rank and spare matrix decomposition (LRaSMD), was further developed in [32] which represents a data matrix $\mathbf{X}$ as a linear decomposition of a low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$ plus a noise matrix $\mathbf{N}$ by

$$\mathbf{X}=\mathbf{L}+\mathbf{S}+\mathbf{N} \quad (3)$$

where an approach called go decomposition (GoDec) developed in [32] can be used to find desired a low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$.

However, the models (1)–(3) only offer separation of these two spaces with anomalies indeed sandwiched between BKG and noise. So, to perform AD effectively, BKG and anomalies must be taken care of simultaneously. When anomalies are strong, they can be embedded in BKG. On the other hand, when anomalies are weak, they can be mistakenly treated as noise, specifically anomalies mixed with noise. However, recent efforts have been devoted to AD enhancement but not BKG removal/suppression. For example, Wang *et al.* [33] proposed an autonomous hyperspectral AD network (Auto-AD) to reconstruct the background by including an adaptive-weighted loss function to assign greater weights to potential anomalous pixels so as to increase the contrast of the anomalies against BKG. Chong *et al.* [34] proposed a hyperspectral deep-net-based anomaly detector using a weight adjustment strategy (WAHyperDNet), which addresses the mixture of anomalous pixels. Li *et al.* [35] modified (2) as the low-rank and sparse decomposition model with a mixture of Gaussians (LSDM-MoG) where the sparse component can be described by a mixture of Gaussian (MoG) noises so that anomalies can be detected by the Manhattan distance (MD). Zhang *et al.* [36] took advantage of the LRaSMD model using the Mahalanobis distance to develop an AD method. Li *et al.* [37] developed a prior-based tensor approximation (PTA) to decompose the data space into a background tensor and an anomaly tensor for hyperspectral AD so that the anomalies can be detected in the anomaly tensor.

This article takes a rather different approach from all the above-mentioned approaches, called component decomposition analysis (CDA) which marriages two well-known component analyses, principal component analysis (PCA) [38] and independent component analysis (ICA) [39], to address both BKG and anomaly issues simultaneously as well as noise, respectively. In analogy with the LRaSMD model in (3), CDA decomposes a data space into principal components (PCs), independent components (ICs), and noise component as a three-component mixture model, called CDA model

$$\mathbf{X}=\mathbf{PCs}+\mathbf{ICs}+\mathbf{N}_{\text{CDA}} \quad (4)$$

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which is produced by orthogonal subspace projection (OSP) [1]. Comparing (4) to (3), the PCs and ICs in (4) can be viewed as the counterparts of **L** and **S**, respectively, in (3) with several key differences. One is that (4) is a statistical model which characterizes the data by statistics of all orders with PCs and ICs specified by the statistics of the first and second orders as BKG, and all other high orders together as signal subspace, respectively. Another is the sums in (4) which are mutual orthogonal compared with the sums of (3) which are not necessarily orthogonal. Accordingly, there is no leakage of BKG specified by PCs into ICs as signal subspace. But this is not true for (3). In order for (3) to be an orthogonal decomposition model, a recent work in [40] derived an OSP-GoDec which made (3) orthogonal. A third one is that CDA uses OSP to annihilate PCs as BKG, while extracting anomalies from ICs.

It should be noted that there are two ways to take care of BKG effects. One is BS which can be performed by inverting the sample covariance/correlation matrix. This is exactly what RX-AD does. But RX-AD does not remove BKG. The other is BKG removal/annihilation which can be performed by OSP. An illustrative example is a recently developed OSP-based AD which took advantage of the LRaSMD model in (3) and used OSP to annihilate the BKG [41].

As for AD enhancement, CDA incorporates the concept of sparsity cardinality (SC) introduced by GoDec into ICs to derive CDASC anomaly detectors (CDASC-ADs). Since we are only interested in anomalies, we can annihilate m PCs completely from the CDA model in (4) by OSP to yield an ICA model given by

$$\hat{\mathbf{X}} = \mathbf{ICs} + \mathbf{N}_{\mathrm{ICA}} \tag{5}$$

and then use SC to enhance anomaly extraction.

Upon implementing models (1)–(3), how to select their model orders and regularization parameters are key issues and have tremendous impacts on the effectiveness of using these models. Unfortunately, these parameters have been determined by the trial-and-error methods empirically. For the proposed CDA model in (4) and ICA model in (5), the number of PCs, m, and the number of ICs, j, can be automatically determined by virtual dimensionality (VD) [1], [42], [43] and minimax-singular value decomposition (MX-SVD) [44], respectively. This advantage is significant compared with the all model-based methods which must fine-tune their parameters.

To evaluate the AD performance, a commonly used measure is the area under a receiver operating characteristic (ROC) curve (AUC) of detection probability $\mathrm{P_D}$ versus false alarm probability, $\mathrm{P_F}$, $\mathrm{AUC_{(D,F)}}$. As noted above, BS plays a key role in AD. However, according to [45], the traditional $\mathrm{AUC_{(D,F)}}$ is not capable of evaluating the BS performance because both $\mathrm{P_D}$ and $\mathrm{P_F}$ are determined by the same threshold. When $\mathrm{P_D}$ is high, $\mathrm{P_F}$ is also very high. Since BS is a critical criterion to measure the effectiveness of AD, the conventional $\mathrm{AUC_{(D,F)}}$ cannot be used to evaluate BS as demonstrated by [45]. More such examples are further discussed in Section VIII. So, to evaluate AD from all aspects, the 3-D ROC-curve-based detection measures derived in [45] are particularly useful and effective for AD performance evaluation.

Several contributions resulting from this article can be summarized as follows.

1) First, a CDA model (4) is introduced so that a CDA theory is therefore derived.
2) To further enhance AD for the CDA model (4), the SC introduced by GoDec is incorporated into the $\mathbf{IC}^j$ component space to constrain the number of potential anomalous target samples to be extracted from the $\mathbf{IC}^j$ data cube. This gives rise to a CDASC model.
3) VD is used to estimate, p, i.e., the total number of PCs and ICs which is the sum of the number of PCs, m, and the number of ICs, j. Once the value of p is determined by VD, j can be estimated by MX-SVD which in turn can derive $m = p - j$.
4) Using the VD-determined m and MX-SVD-determined j, the CDA algorithm and CDASC-AD are developed.
5) Comparative studies and analyses are conducted in Section VIII to specifically demonstrate significant advantages offered by 3-D ROC-curve-derived detection measures in AD performance evaluation in terms of anomaly detectability, BS, joint anomaly detectability-BS, and overall detection performance.

The remainder of this article is organized as follows. Section II is particularly devoted to the development of CDA theory which includes the CDA model and CDA algorithm. Section III develops the CDASC theory which includes introduction of SC and development of the CDASC algorithm. After CDA and CDASC theories are introduced, Section IV describes VD/MX-SVD which automatically determines the values of the parameters m and j. Once CDA/CDASC and VD/MX-SVD are derived, Section V uses CDA/CDASC to design various RX/R-AD for AD. Section VI includes 3-D-ROC curve-derived detection measures for AD performance analysis. Section VII describes three real images to be used for experiments and presents the experimental results and discussion. Section VIII conducts a comparative analysis with two current state-of-the-art AD methods along with discussions. Section IX provides a brief conclusion.

II. CDA Theory

Component analysis has been widely used in many applications such as data compression, data representation, and data preprocessing where two types of component analysis are of particular interest, PCA and ICA, each of which has its own specific applications. For example, PCA has found a wide range of applications in remote sensing [46]. It uses eigenvalues to decompose data samples into a set of PCs specified by eigenvectors which retain most of the data information.

A. PCA Model

According to the factor analysis in [47] and [48], all the eigenvalues can be categorized into three sets of eigenvalues, m primary eigenvalues corresponding to m largest eigenvalues, j secondary eigenvalues corresponding to the next j largest eigenvalues, and the remaining eigenvalues corresponding to

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noise. PCA uses these three sets of eigenvalues to find three sets of eigenvectors, $\{\mathbf{v}_l\}_{l=1}^{m}$, $\{\mathbf{v}_{m+l}\}_{l=1}^{j}$, and $\{\mathbf{v}_l\}_{l=m+j+1}^{L}$ to generate three components, PCs, minor components (MCs), and noise components, respectively [49], to be defined as the PCA model in this article as

$$\mathbf{X} = \mathbf{PC}^m + \mathbf{MC}^j + \mathbf{N}_{\mathrm{PCA}} \tag{6}$$

where PCs, MCs, and noise components can be used to specify BKG, anomalies, and noise, respectively.

B. ICA Model

On the other hand, ICA [39] was originally developed for blind source separation and has found great applications in speech recognition and functional magnetic resonance imaging. It makes use of high-order statistics to unmix signals as blind sources which are separated in different ICs. Most importantly, unlike the works in [4], [19], [28], and [35] which made a key assumption that noise was Gaussian-distributed, the separated noise by ICA is non-Gaussian, a scenario which is generally true in hyperspectral imagery. To implement ICA, a mandatory preprocessing data sphering (DS) must be performed to remove the statistics of the first and second orders to produce sphered data, $\tilde{\mathbf{X}}$. Then we can define a j-IC component ICA model as

$$\tilde{\mathbf{X}} = \mathbf{IC}^j + \mathbf{N}_{\mathrm{ICA}}. \tag{7}$$

Since PCA is a transformation of second-order statistics, it can be used to characterize BKG effectively via a set of only PCs. It is known that signals are generally considered as outliners and can be characterized more effectively by higher orders of statistics. This is crucial for anomalies. ICA turns out to be a perfect candidate that meets what we need. In particular, it removes the first- and second-order statistics and uses its ICs to specify signal sources to perform blind source separation. One of the major motivations is "Can we combine models (6) and (7) and merge into a single model so that PCA-generated eigenvectors can be used to remove BKG, while ICA can be used to detect anomalies?" This seems easier said than done. One issue is that the PCA-generated eigenvectors are not the same as ICA-generated projection vectors. Simply merging these two sets of vectors does not work. Interestingly, it looks like that a mixed PCA/ICA developed in [50] provides a feasible answer.

C. CDA Model

The CDA model intends to integrate the PCA model in (6) and the ICA model in (7) into a single model by replacing the $\mathbf{MC}^j$ and $\mathbf{N}_{\mathrm{PCA}}$ components in (6) with $\mathbf{IC}^j$ and $\mathbf{N}_{\mathrm{ICA}}$ in (7). Since the eigenvectors $\{\mathbf{v}_l\}_{l=1}^{j}$ generated by PCA cannot be used for ICA to generate ICs, we must find an OSP projector $P_{\mathbf{PC}^m}^{\perp}$ defined by

$$P_{\mathbf{PC}^m}^{\perp} = \mathbf{I} - \left(\mathbf{PC}^m\right)^T \left(\mathbf{PC}^m \left(\mathbf{PC}^m\right)^T\right)^{-1} \mathbf{PC}^m \tag{8}$$

to project the original dataset $\mathbf{X}$ into its orthogonal complement subspace of $\mathbf{PC}^m$. Then its resultant subspace, $P_{\mathbf{PC}^m}^{\perp}\mathbf{X}$, can be used to generate our desired set of j IC projection vectors $\{\mathbf{u}_l\}_{l=1}^{j}$. To do so, FastICA [51] is used to form the $\mathbf{IC}^j$ component space. In this case, both $\mathbf{PC}^m$ and $\mathbf{IC}^j$ are orthogonal each other. As a result, combining $\{\mathbf{v}_l\}_{l=1}^{m}$ and $\{\mathbf{u}_l\}_{l=1}^{j}$ creates a hybrid set of orthonormal basis vectors $\{\hat{\mathbf{v}}_l\}_{l=1}^{m+j}$ defined by

$$\hat{\mathbf{v}}_l = \begin{cases} \mathbf{v}_l & \text{for } 1 \le l \le m \\ \mathbf{u}_{s+m} & \text{for } 1 \le s \le j. \end{cases} \tag{9}$$

By virtue of (9), the original data space can be represented by $\mathbf{PC}^m = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_m]$ and $\mathbf{IC}^i = \left[\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_j\right]$ components as

$$\mathbf{X} = \mathbf{PC}^m + \mathbf{IC}^i + \mathbf{N}_{\mathrm{CDA}}. \tag{10}$$

Comparing (10) to (3) again, an immediate finding is that $\mathbf{PC}^m$ and $\mathbf{IC}^j$ can also be considered to correspond to the low-rank matrix $\mathbf{L}$ and the sparse matrix $\mathbf{S}$, respectively.

Once the CDA model in (10) is established, the next key issue that needs to be resolved is how to reliably estimate or determine the values of m and j. This issue is similar to determination of model orders and regularization parameters encountered in all the models specified by (1)–(3). Fortunately, the concepts of VD and MX-SVD in [40] can be used to determine appropriate values of m and j in (10) and will be discussed in detail in Section IV.

Algorithm 1 CDA Algorithm

1. Initial conditions: m and j are given
2. Use a sequential PCA (SQ-PCA) developed in [52,53] to find the first m eigenvectors, $\{\mathbf{v}_l\}_{l=1}^{m}$.
3. Use $\{\mathbf{v}_l\}_{l=1}^{m}$ in (9) to find $\mathbf{X}_{\mathbf{V}_m^{\mathrm{PCA}}} = P_{\mathbf{PC}^m}\mathbf{X}$ where
$$P_{\mathbf{PC}^m} = \left(\mathbf{PC}^m\right)^T \left(\mathbf{PC}^m \left(\mathbf{PC}^m\right)^T\right)^{-1} \mathbf{PC}^m \tag{11}$$
4. Use (8) to find $\mathbf{X}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp} = P_{\mathbf{PC}^m}^{\perp}\mathbf{X}$
$$P_{\mathbf{PC}^m}^{\perp} = \mathbf{I} - \left(\mathbf{PC}^m\right)^T \left(\mathbf{PC}^m \left(\mathbf{PC}^m\right)^T\right)^{-1} \mathbf{PC}^m. \tag{12}$$
5. Find the sphered $\tilde{\mathbf{X}}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp}$ in (7).
6. Apply FastICA to $\tilde{\mathbf{X}}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp}$ to find the first j IC projection vectors $\{\mathbf{u}_s\}_{s=1}^{j}$ and form
$$\mathbf{U}_j^{\mathrm{ICA}} = \left[\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_j\right]. \tag{13}$$
It should be noted that since FastICA uses random initial conditions to generate ICs, the produced ICs are not consistent. To resolve such a random issue, ICA-DR3 developed in [54] is used here to produce consistent ICs due to its use of automatic target generation process (ATGP)-generated targets developed in [55] as initial conditions. In addition, when FastICA is implemented in the CDA algorithm, a sphering of $\mathbf{X}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp}$ is also performed prior to applying FastICA and its resulting sphered is denoted by $\tilde{\mathbf{X}}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp}$.
7. Find
$$\mathbf{IC}_{\mathrm{CDA}}^j = \mathbf{X}_{\mathbf{U}_j^{\mathrm{ICA}}} = P_{\mathbf{U}_j^{\mathrm{ICA}}} \tilde{\mathbf{X}}_{\mathbf{V}_m^{\mathrm{PCA}}}^{\perp}. \tag{14}$$

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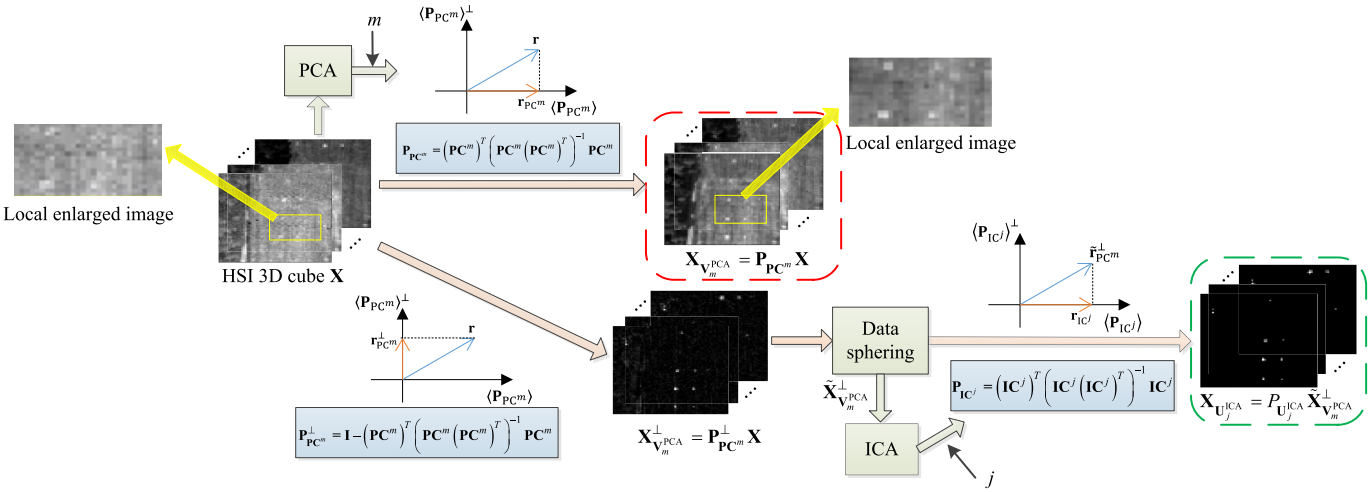


Fig. 1. Graphic diagram of the step-by-step implementation of the CDA algorithm.

D. CDA Algorithm

From a statistical point of view, PCA is a second-order statistics transformation which finds eigenvalues to generate eigenvectors to represent the data in terms of the magnitude of variances. Accordingly, the first few PCs generated by the eigenvectors corresponding to the first few largest eigenvalues generally represent BKG. On the other hand, ICA is a high-order statistics transformation which uses a criterion of high-order statistics greater than 2 to find IC projection vectors which can be used to characterize non-Gaussian distributed data. Consequently, the ICs generated by ICA are generally used to separate unknown non-Gaussian signal sources from Gaussian-distributed data. Using the properties of PCA and ICA, we can design an algorithm to find PCs and ICs for the CDA model in (10) as shown in Algorithm 1.

It should be noted that unlike ICA which separates signal sources in j different ICs, CDA constructs a j-component IC space $\mathbf{U}_j^{\mathrm{ICA}}$ which retains anomalies. Using the above CDA algorithm, we can construct the $\mathbf{PC}^m + \mathbf{IC}^j + \mathbf{N}$ model in (10). Fig. 1 provides a detailed graphic diagram to illustrate step-by-step implementations of the CDA algorithm.

As a final comment, combining PCs and ICs to perform data analysis is not new. For example, Chai *et al.* [50] developed a mixed PCA/ICA approach to spectral data compression and [56] took advantage of PCs and ICs to find endmembers to perform spectral unmixing. However, what is new is how to produce $\mathbf{PC}^m$ and $\mathbf{IC}^j$ for the CDA model in (10). The proposed CDA algorithm is particularly designed to meet this need. Specifically, the ICA used in (10) is different from the ICA used in [50] and [56] which was applied to the original data matrix $\mathbf{X}$ not same as $\mathbf{X}^{\perp}_{\mathbf{V}_m^{\mathrm{PCA}}}$ obtained by (13) using the CDA model.

A relationship between the PCA model in (6) and the CDA model in (10) is depicted in Fig. 2 where $\mathbf{r}_{\mathbf{PC}^m}$, $\mathbf{r}_{\mathbf{MC}^j}$, and $\mathbf{r}_{\mathbf{IC}^j}$ are the orthogonal projections of a data sample vector $\mathbf{r}$ mapped on the $\mathbf{PC}^m$, $\mathbf{MC}^j$, and $\mathbf{IC}^j$ components, respectively.

The novelties of CDA are: 1) the development of the CDA model in (10) to integrate PCA and ICA and 2) the design of a CDA algorithm to decompose $\mathbf{PC}^m$ and $\mathbf{IC}^j$ plus noise into the CDA model.

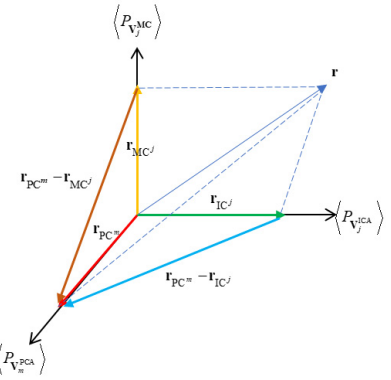


Fig. 2. Relationship between PCA and CDA models where $\mathbf{r}_{\mathbf{PC}^m}$, $\mathbf{r}_{\mathbf{MC}^j}$, $\mathbf{r}_{\mathbf{IC}^j}$, $\mathbf{r}_{\mathbf{IC}^j}$ are orthogonal projections of a data sample vector r mapped on the $\mathbf{PC}^m$, $\mathbf{MC}^j$, $\mathbf{IC}^j$ components respectively.

III. CDASC Theory

Despite that CDA can separate noise from $\mathbf{IC}^j$, the $\mathbf{IC}^j$ data cube may still contain certain noisy samples. It is crucial to remove these noisy samples from this $\mathbf{IC}^j$ space to enhance AD. Interestingly, the concept of SC used by GoDec in [32] offers a practical solution.

A. Sparsity Cardinality

SC is quite different from the sparsity order commonly used by LRaSR, which is generally obtained by regularization-constrained optimization problems through Lagrange multipliers or alternating direction method of multipliers (ADMM). SC originates from video processing to detect anomalies. It considers individual frame as a single image and a set of L image frames as an L-dimensional image data cube. So, SC $= k$ is defined as the number of the first k largest nonzero pixel values among all the individual image frames. To translate SC to hyperspectral imagery, a single image frame is considered as a spectral band and a set of L image frames

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Algorithm 2 CDASC

Steps 1-7 are the same as CDA.

8. Determine k for SC.
   Let $k = j \times N$ be the data sparsity cardinality.
   Calculate the sparse space in $\mathbf{IC}^j_{\mathrm{CDA}}$, $\mathbf{IC}^j_{\mathrm{CDASC}} = P_\Omega\left(\mathbf{IC}^j_{\mathrm{CDA}}\right)$ where $P_\Omega(\cdot)$ is the projection of a matrix onto an entry set Ω which is a subset containing the first k largest nonzero pixel values in of $\mathbf{IC}^j_{\mathrm{CDA}}$, defined by

$$\mathbf{IC}^j_{\mathrm{CDA}} = \left\{\mathbf{r}_{r,c} = (r^1_{r,c}, r^2_{r,c}, \ldots, r^L_{r,c})^T\right\} \tag{17}$$

   where $r^l_{r,c} = \begin{cases} r^l_{r,c}; & \text{if } r^l_{r,c} \in \Omega \\ r^l_{r,c} = 0; & \text{otherwise.} \end{cases}$

is an L-spectral band image data cube. Therefore, SC $= k$ can be interpreted as the number of k largest nonzero pixel values across all L individual spectral band images. Imposing SC $= k$ on the $\mathbf{IC}^j$ data cube, we are able to make it a sparse space constrained by SC, denoted by $\mathbf{IC}^j_{\mathrm{SC}}$. The key idea behind SC is based on the fact that a noise pixel value is generally very small and cannot be among the set of the k largest nonzero pixel values if k is appropriately determined. Using SC as a constraint, the CDA model in (10) can be further extended to a CDASC model specified by

$$\mathbf{X} = \mathbf{PC}^m + \mathbf{IC}^j_{\mathrm{SC}} + \mathbf{N}_{\mathrm{CDASC}}. \tag{15}$$

B. Development of CDASC Algorithm

Due to the fact that ICA does not discriminate signal sources from non-Gaussian noise, $\mathbf{X}_{\mathbf{U}^{\mathrm{ICA}}_j} = P_{\mathbf{U}^{\mathrm{ICA}}_j}\tilde{\mathbf{X}}^{\perp}_{\mathbf{V}^{\mathrm{PCA}}_m}$ obtained by (14) may contain unwanted noisy samples. However, noise samples are generally not sparse in the sense that the values of its spectral components are relatively small but not zeros. As a consequence of imposing SC on $\mathbf{X}_{\mathbf{U}^{\mathrm{ICA}}_j}$, the resulting $\mathbf{IC}^j_{\mathrm{SC}}$ in the CDASC model in (15) can further reduce the noise effect.

According to [32], GoDec is designed to find the low-rank matrix $\mathbf{L}$ and the sparse matrix $\mathbf{S}$ in (3) by minimizing the following constrained optimization decomposition error:

$$\min_{\mathbf{L},\mathbf{S}} ||\mathbf{X} - \mathbf{L} - \mathbf{S}||^2_F \tag{16}$$

subject to $rank(\mathbf{L}) \le m$ and card$(\mathbf{S}) \le k$ where $||\cdot||_F$ is the Frobenius norm, and m and k must be specified *a priori*.

By taking advantage of SC, we can also impose SC $= k$ on $\mathbf{IC}^j_{\mathrm{CDA}} = \mathbf{X}_{\mathbf{U}^{\mathrm{ICA}}_j}$ in (14) using a similar manner that SC is imposed on the sparse matrix $\mathbf{S}$ in GoDec to derive a new CDASC algorithm for (15), which includes SC $= k$ in CDA as an addition step, step 8 described in Algorithm 2. The resulting $\delta^{\text{RX-AD}}_{\mathrm{CDA}}(\mathbf{r})$ becomes $\delta^{\text{RX-AD}}_{\mathrm{CDASC}}(\mathbf{r})$.

Fig. 3 particularly shows how SC in step 8 of CDASC works. For a given $\mathbf{IC}^j_{\mathrm{CDA}}$ shown in Fig. 3(a), it shows 3×3 windows in the spectral bands, denoted by the first band, $\mathbf{B}_1$, the lth band, $\mathbf{B}_l$, and the Lth band, $\mathbf{B}_L$ whose values are specified by $x_{i,s}$ with $i = 1, l, L$ and $s \in \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ in Fig. 3(b). As defined in (17), the values in the three 3×3 windows in the spectral bands, $\mathbf{B}_1$, $\mathbf{B}_l$, and $\mathbf{B}_L$, after SC become

$$x_{i,s} = \begin{cases} x_{i,s}; & \text{if } x_{i,s} \in \tau \\ 0; & \text{if } x_{i,s} \notin \tau \end{cases}$$

as shown in Fig. 3(c) where only two values in the 3×3 window of the first band have values $\ge \tau$, only five values in the 3×3 window of the lth band have values $\ge \tau$, and only three values in the 3×3 window of the first band have values $\ge \tau$. The final $\mathbf{IC}^j_{\mathrm{CDASC}}$ is shown in Fig. 3(d).

It is worth noting that the order of sparsity, j, which is the rank of a sparse matrix, is very different from SC $= k$, which is defined as the number of nonzero signal components. However, since there are N data sample vectors and each data sample vector has sparsity of order j, SC can be determined by $k = j \times N$ [36].

The novelties of CDASC are: 1) the development of the CDASC model in (17) by imposing SC on CDA and 2) the design of a CDASC algorithm to realize the CDASC model. The evolution to the CDASC model can be described by

$$\left.\begin{array}{l}\text{PCA model (6)}\\ \text{ICA model (7)}\end{array}\right\} \Rightarrow \text{CDA model (10)} \Rightarrow \text{CDASC model.}$$

IV. Determination of Numbers of PCs and ICs

In order for CDA to work effectively, finding appropriate values of m and j in the CDA model (10) is crucial and critical to its success.

Instead of directly determining the values of m and j for (10), we take a rather different approach by estimating the sum of m and j, denoted by $p = m + j$. It has been shown in [1], [42], and [43] that VD can be used to estimate the number of spectral signatures in hyperspectral imagery where such spectral signatures should include BKG signatures and each target signal signatures. So, if each BKG signature and each target signature can be interpreted to specify one PC and one IC, respectively, the value of VD, n_{VD}, denoted by p should be the sum of m (i.e., number of PCs) and j (i.e., number of ICs), i.e., $n_{\mathrm{VD}} = p = m + j$. The method to use to estimate n_{VD} is the one developed by Harsanyi–Farrand–Chang (HFC) [57], commonly referred to as the HFC method along with its MATLAB codes also available in [38, pp. 997–998]. It considers the differences $\{z_l\}^L_{l=1} = \left\{\bar{\lambda}_l - \lambda_l\right\}^L_{l=1}$ between the eigenvalues $\left\{\bar{\lambda}_l\right\}^L_{l=1}$ of the correlation matrix $\mathbf{R}$ and the eigenvalues $\{\lambda_l\}^L_{l=1}$ of the covariance matrix $\mathbf{K}$ as signal sources. If there is a signal present in the lth component for any l, then this signal will contribute a mass as a mean to make $\bar{\lambda}_l - \lambda_l > 0$. Otherwise, $\bar{\lambda}_l - \lambda_l = 0$ if there is no signal present in the lth component. With this interpretation, the HFC method formulates the VD estimation as a binary simple composite hypothesis testing problem as follows:

$$\begin{aligned} &\mathrm{H}_0(\text{signature absence}) : p_0(z_l) \approx \bar{\lambda}_l - \lambda_l = 0 \\ &\text{versus} \\ &\mathrm{H}_1(\text{signature presence}) : p_1(z_l) \approx \bar{\lambda}_l - \lambda_l > 0. \end{aligned} \tag{18}$$

A Neyman–Pearson detector of $z_l = \bar{\lambda}_l - \lambda_l$, denoted by $\delta^{\mathrm{NP}}(z_l)$ for (18), can be obtained by maximizing the detection power PD subject to the false alarm probability PF being fixed

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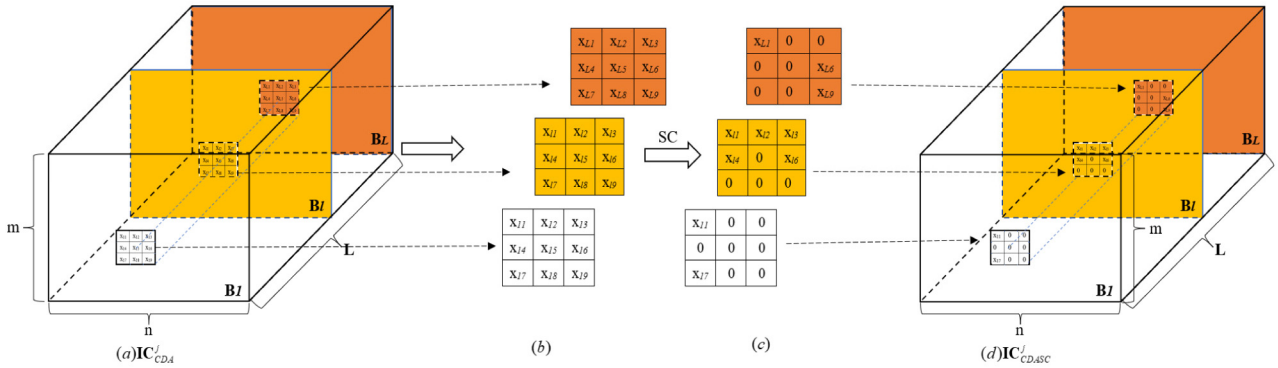


Fig. 3. Illustrative diagram of how SC works. (a) $\mathbf{IC}^j_{\mathrm{CDA}}$, (b) 3×3 windows in the three selected spectral bands, $\mathbf{B}_1$, $\mathbf{B}_l$ and $\mathbf{B}_L$, (c) 3×3 windows in the spectral bands, $\mathbf{B}_1$, $\mathbf{B}_l$ and $\mathbf{B}_L$ after SC, (d) $\mathbf{IC}^j_{\mathrm{CDASC}}$.

at a given value, β, which is the level of significance for the detector [58]. This β is then used to determine the threshold value τ_l in the following randomized decision rule:

$$\delta^{\mathrm{NP}}(z_l) = \begin{cases} 1; & \text{if } \Lambda(z_l) > \tau_l \\ 1 \text{ with probability } \kappa; & \text{if } \Lambda(z_l) = \tau_l \\ 0; & \text{if } \Lambda(z_l) < \tau_l \end{cases} \tag{19}$$

where the likelihood ratio test $\Lambda(z_l)$ is given by $\Lambda(z_l) = p_1(z_l)/p_0(z_l)$ with $p_0(z_l)$ and $p_1(z_l)$ being the probability distributions under H_0 and H_1 for (18), respectively. So, a case of $z_l = \hat{\lambda}_l - \lambda_l > \tau_l$ indicates that $\delta^{\mathrm{NP}}(z_l)$ in (19) fails the test, which means that there is signal energy assumed to contribute to the eigenvalue, $\hat{\lambda}_l$, in the lth spectral band. It should be noted that the test (19) must be performed for each L spectral band. Therefore, for each pair of $\hat{\lambda}_l - \lambda_l$, the threshold τ_l varies with the fixed value of β and is spectral-band-dependent. Using (19), n_{VD} can be determined by calculating

$$n^{\mathrm{HFC}}_{\mathrm{VD}}(\mathrm{P_F} \le \beta) = \sum_{l=1}^{L} \lfloor \delta^{\mathrm{NP}}(z_l) \rfloor \tag{20}$$

with a predetermined false alarm probability $\mathrm{P_F} = \beta$ where $\lfloor \delta(z_l) \rfloor = 1$ only if $\delta^{\mathrm{NP}}(z_l) = 1$ and $\lfloor \delta(z_l) \rfloor = 0$ if $\delta^{\mathrm{NP}}(z_l) < 1$. A noise-whitened HFC (NWHFC) method was later developed which whitens the data prior to the HFC method. The resulting n_{VD} is defined as $n^{\mathrm{NWHFC}}_{\mathrm{VD}}(\mathrm{P_F} \le \beta)$. More details about VD can be found in [43].

Now a follow-up issue is how to break up p into m and j because VD does not provide any means of distinguishing a signal source from a BKG source. Fortunately, Kuybeda *et al.* [44] offer a solution which developed a maximum orthogonal complement algorithm (MOCA) to estimate the value of p and then the value of j by MX-SVD. Subtracting j from p gives rise to $m = p - j$. This approach was first developed in [40] to estimate the ranks of $\mathbf{L}$ and $\mathbf{S}$ used for the LRaSMD model in (4). More details can be found in [40].

The novelty here is to use VD and MX-SVD to determine the number of PCs, m in the $\mathbf{PC}^m$ component space and the number of ICs and j in the $\mathbf{IC}^j$ component space.

V. CDA AND CDASC FOR AD

Since RX-AD was developed in 1990, many of its variants and extensions have been developed and reported in the literature. Basically, there are two types of anomaly detectors which are commonly used. One is called RX-AD developed by Reed and Yu [4], which makes use of the global sample auto-covariance matrix $\mathbf{K}$ and global sample mean μ to perform AD on the data sample vector $\mathbf{r}$ given by

$$\delta^{\text{RX-AD}}(\mathbf{r}) = (\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{r} - \boldsymbol{\mu}). \tag{21}$$

The other is called R-AD developed in [1] and [5], which makes use of the global sample auto-correlation matrix $\mathbf{R}$ to perform AD on the data sample vector $\mathbf{r}$ given by

$$\delta^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r}. \tag{22}$$

To perform RX/R-AD in (21) and (22) effectively, Chang [1] and Chang and Chiang [5] separated RX-AD in (21) and R-AD in (22) into two different operations, a BS/removal process using $\mathbf{K}^{-1}(\mathbf{r} - \mu)/\mathbf{R}^{-1}\mathbf{r}$ and a matched filter using $(\mathbf{r} - \mu)/\mathbf{r}$ as a matching signal. By means of this separation, instead of performing (21) and (22) on the same original data space $\mathbf{X}$, we can perform RX/R-AD via the CDA model in (10) by implementing the BS/removal process in one space, $\mathbf{PC}^m$ component space, and a match filtering in another space, $\mathbf{IC}^j$ component space.

Upon implementing the CDA model for AD, two key parameters, which have tremendous impacts on the performance of AD, are the sample covariance matrix $\mathbf{K}$ in (21)/the sample correlation matrix $\mathbf{R}$ in (22) and the data sample vector $\mathbf{r}$. Because the role of inverting $\mathbf{K}/\mathbf{R}$ in (21) and (22) is to suppress the effects of data sample vectors embraced by $\mathbf{K}/\mathbf{R}$, the data samples to be included in $\mathbf{K}/\mathbf{R}$ should be all BKG signal samples and not potential anomalies. Furthermore, $\mathbf{PC}^m$ in (6) and (10) is generally used to characterize BKG which is unlikely to contain anomalies. So, in this case, $\mathbf{PC}^m$ can be suppressed by inverting $\mathbf{K}/\mathbf{R}$. On the other hand, in order for the data sample vector $\mathbf{r}$ to be extracted as a potential anomaly, the space on which $\mathbf{r}$ is operated should not be BKG samples but rather signal samples, which is $\mathbf{IC}^j$ used to stand for $\mathbf{IC}^j_{\mathrm{CDA}}$ for simplicity. With this in mind, (21) and (22) can be modified as

$$\delta^{\text{RX-AD}}_{\mathbf{B}}(\mathbf{r_A}) = (\mathbf{r_A} - \boldsymbol{\mu}_{\mathbf{A}})^T \mathbf{K}^{-1}_{\mathbf{B}} (\mathbf{r_A} - \boldsymbol{\mu}_{\mathbf{A}}) \tag{23}$$

and

$$\delta^{\text{R-AD}}_{\mathbf{B}}(\mathbf{r_A}) = \mathbf{r}^T_{\mathbf{A}} \mathbf{R}^{-1}_{\mathbf{B}} \mathbf{r_A} \tag{24}$$

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where the subscripts "**A**" and "**B**," referred to as anomaly space and BKG space, respectively, are used to indicate the components on which **r** and **K/R** are operated, respectively. There are two different scenarios, $\mathbf{PC}^m$ and $\mathbf{PC}^m+\mathbf{IC}^j$ that can be used to characterize BKG, while another two scenarios $\mathbf{IC}^j$ and $\mathbf{PC}^m+\mathbf{IC}^j$ can also be used to specify signal sources as anomalies. Consequently, a total of six scenarios are of interest and used for experiments. $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$, $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m}+\mathbf{r}_{\mathbf{IC}^j})$, $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m}+\mathbf{r}_{\mathbf{IC}^j})$, $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$, $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$, and $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ where the superscript "RX/R-AD" indicates either RX-AD or R-AD.

The novel contribution is to use the CDA and CDASC models to extend RX-AD and R-AD to various anomaly detectors which can be operated in two different separate spaces, BKG space for removal and signal space for anomaly extraction with $\mathbf{PC}^m$ and $\mathbf{IC}^j$ in the CDA and CDASC models.

VI. 3-D ROC-Curve-Derived Detection Measures

The 2-D ROC curve has been widely used to measure the performance of AD. It is a plot of detection probability P_D versus false alarm probability P_F. Its area under curve (AUC), denoted by $AUC_{(D,F)}$, is calculated as an indicator to measure how effective a detector is. Unfortunately, according to recent reports in [3], [41], [45], and [59], solely relying on $AUC_{(D,F)}$ is not creditable. This is because both P_D and P_F are calculated by the same threshold τ used by the detector. As a result, when both P_D and P_F are very high, its calculated $AUC_{(D,F)}$ is also very high. Similarly, when both P_D and P_F are very low, its calculated $AUC_{(D,F)}$ is also very low. In other words, P_D and P_F are tied together simultaneously and cannot work independently. Thus, P_D and P_F cannot be used individually to measure target detectability (TD) and BS, respectively. To address this issue, Chang [45] developed an effective 3-D ROC-analysis-based evaluation tool, which extends the traditional 2-D ROC analysis by including the threshold τ as an additional independent parameter to represent a 3-D ROC curve as a function of three parameters, P_D, P_F, and τ. As a result, a 3-D ROC curve can be generated by a triplet parameter vector specified by (P_D,P_F,τ). This 3-D ROC curve can be further used to generate three 2-D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) with their respective AUC values, denoted by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$. The values of $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ can be used to evaluate TD and BS, respectively. In addition to these three AUC values, five new AUC measures were also developed in [3], [41], [45], and [59] to measure TD, BS, joint TD and BS (TDBS), signal-to-noise probability ratio (SNPR), and overall detection probability (ODP). Specifically,

1) $AUC_{(D,F)}$: Effectiveness of a detector.
2) $AUC_{(D,\tau)}$: TD of a detector.
3) $AUC_{(F,\tau)}$: BS of a detector.
4) AUC_{TD} : Joint TD of a detector defined by

$$0 \le \text{AUC}_{\text{TD}} = \text{AUC}_{(\text{D,F})} + \text{AUC}_{(\text{D},\tau)}) \le 2. \quad (25)$$

5) AUC_{BS} : Joint BS of a detector defined by

$$-1 \le \text{AUC}_{\text{BS}} = \text{AUC}_{(\text{D,F})} - \text{AUC}_{(\text{F},\tau)}) \le 1. \quad (26)$$

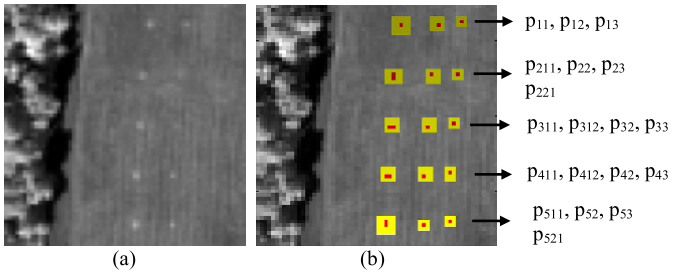


Fig. 4. (a) HYDICE15-panel scene which contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels.

6) AUC_{SNPR} : SNPR of a detector defined by

$$0 \le \text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(\text{D},\tau)}}{\text{AUC}_{(\text{F},\tau)}}. \quad (27)$$

7) AUC_{TDBS} : TDBS of a detector defined by

$$-1 \le \text{AUC}_{\text{TDBS}} = \text{AUC}_{(\text{D},\tau)} - \text{AUC}_{(\text{F},\tau)}) \le 1. \quad (28)$$

8) AUC_{ODP} : ODP of a detector defined by

$$-1 \le \text{AUC}_{\text{ODP}} = \text{AUC}_{(\text{D,F})} + \text{AUC}_{(\text{D},\tau)} - \text{AUC}_{(\text{F},\tau)} \le 2. \quad (29)$$

The novelties of the above detection measures can be summarized in accordance with their functionalities described as follows.

1) *TD:* $AUC_{(D,\tau)}$, AUC_{TD}.
2) *BS:* $AUC_{(F,\tau)}$, AUC_{BS}, AUC_{SNPR}.
3) *Overall Performance:* $AUC_{(D,F)}$, AUC_{TDBS}, AUC_{ODP}.

For those who are interested in using the 3-D ROC analysis developed in [45] for detection performance evaluation, a software package of calculating the above detection measures is available on the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL) at University of Maryland, Baltimore County (UMBC), https://wiki.umbc.edu/display/rssipl/10.+Download.

VII. Real Image Experiments and Discussions

Three hyperspectral real image scenes were used for experiments.

A. Real Hyperspectral Images Used for Experiments

1) HYDICE 15-Panel Scene: An airborne hyperspectral digital imagery collection experiment (HYDICE) scene is as shown in Fig. 4. It was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling distance approximately 1.56 m. This scene has been studied extensively by many reports such as [1], [2], and [38]. A total of 169 bands were used for the experiments with low signal/high noise bands: bands 1–3 and bands 202–210; and water vapor absorption bands: bands 101–112 and bands 137–153 were removed. There are 15 square panels in Fig. 4(a) with three different sizes, 3 m×3 m, 2 m×2 m, and 1 m×1 m, respectively. Due to the ground sampling distance approximately 1.56 m, each of the panel in the first column except the first row contains two

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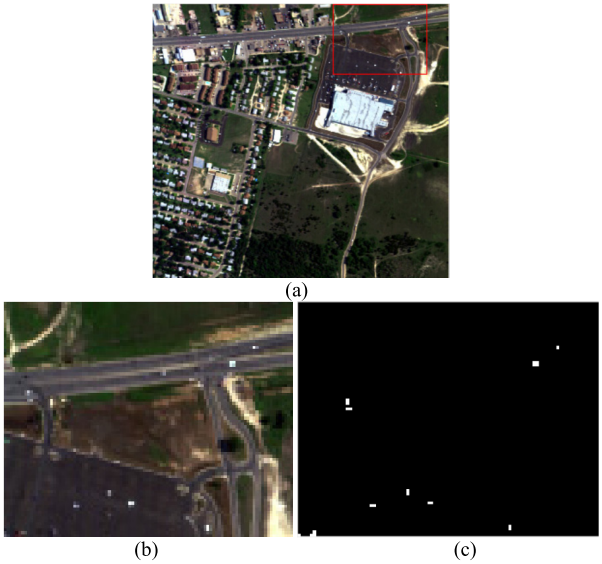


Fig. 5. HYDICE urban scene. Pseudo-color image of (a) whole scene and (b) selected area. (c) Ground = truth map.

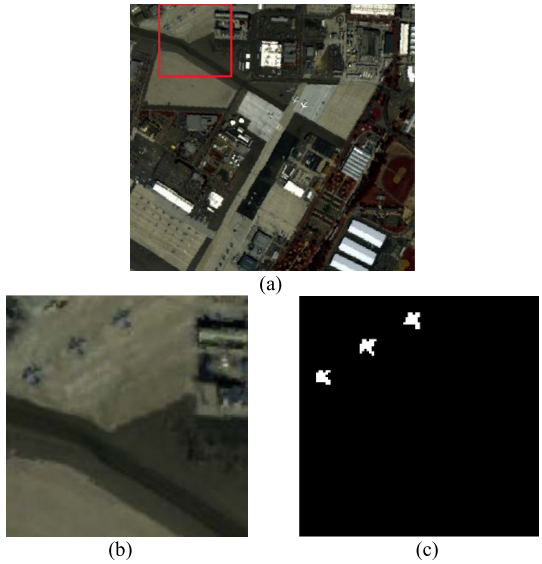


Fig. 6. San Diego Airport scene. Pseudo-color image of (a) whole scene and (b) selected area. (c) Ground-truth map.

panel pixels highlighted by red, p_{211}, p_{221} in row 2, p_{311}, p_{312} in row 3, p_{411}, p_{412} in row 4, and p_{511}, p_{521} in row 5, as shown in Fig. 4. All the remaining 11 panels in Fig. 4 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, p_{13} in row 1, p_{22}, p_{23} in row 2, p_{32}, p_{33} in row 3, p_{42}, p_{43} in row 4, and p_{52}, p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 4(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG.

2) HYDICE Urban Scene: A second dataset is another HYDICE image with pseudo-color as shown in Fig. 5(a) which is available and can be downloaded from the Remote Sensing Laboratory website at https://rslab.ut.ac.ir/data. It is an urban scene and comprises 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 × 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 5(b) along with the ground-truth map shown in Fig. 5(c) where 21 pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from BKG.

3) San Diego Airport Data: A third dataset used for experiments is available in [60] and was also an airborne visible/infrared imaging spectrometer (AVIRIS) acquired scene, which is a San Diego airport scene in CA, USA, with pseudo-color as shown in Fig. 6(a). It is an urban scene in which the main background materials are roof, shadow, and grass. It has a size of 400 × 400 pixels with a 3.5 m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the water absorption regions, low signal-to-noise ratio, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 available bands of the data were retained in the experiments. An area of 100 × 100 pixels at the upper left corner of the scene was selected as the test image shown in Fig. 6(b) along with its ground-truth map shown in Fig. 6(c).

TABLE I
m ESTIMATED FOR HYDICE DATA BY HFC/NWHFC USING MX-SVD

	p	*m by MX-SVD*
HYDICE 15-panel scene	9/13	5/7
HYDICE urban scene	9/13	5/7
San Diego airport	10/11	2/2

There are three airplanes in the image, which consist of 85 pixels and account for 0.85% of the test image as shown in Fig. 6(b). This scene is less interesting than the HYDICE 15-panel data in Fig. 4 because only ground truth is available in three airplanes which are large targets.

Table I shows the value of p estimated by the HFC and HFC/NWHFC methods and the value of m estimated by MX-SVD for all the three scenes.

B. Experiments and Discussions

To evaluate the performance of CDASC, a sparse representation-based model, referred to as l_2-norm minimization and distance weighted regularization matrix and sum-to-one constraint (CRD-DW-STO), developed in [12], and an OSP method, referred to an OSP-based anomaly detector with DS (OSPDS-AD), developed in [41], were used for comparison since both of them have shown to perform very well in the AD compared with other AD methods. In addition, classic RX-AD developed by Reed and Yu [4] and R-AD were also included as a benchmark comparison.

Six versions of CDASC anomaly detectors derived from (21) to (22), which are $\delta_{\mathbf{IC}^j}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{IC}^j})$, $\delta_{\mathbf{PC}^m}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{IC}^j})$, $\delta_{\mathbf{PC}^m+\mathbf{IC}^j}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{IC}^j})$, $\delta_{\mathbf{IC}^j}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$, $\delta_{\mathbf{PC}^m}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$, and $\delta_{\mathbf{PC}^m+\mathbf{IC}^j}^{\text{RX-AD/R-AD}}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$, were used for experiments. It should be noted that when $\mathbf{IC}_{\text{CDASC}}^j$ was implemented for CDASC, we use the notation of $\mathbf{IC}^j$ to stand for $\mathbf{IC}_{\text{CDASC}}^j$ as its abbreviation for simplicity.

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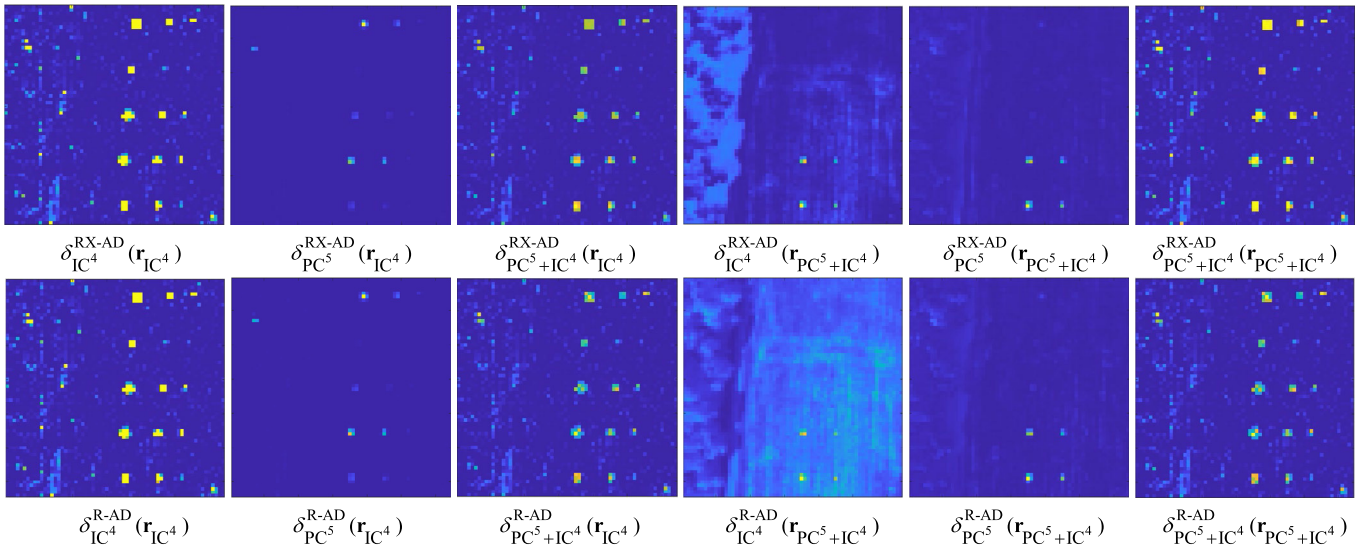


Fig. 7. Detection maps of the HYDICE 15-panel scene using CDASC with $p = 9$, $m = 5$, and $j = 4$.

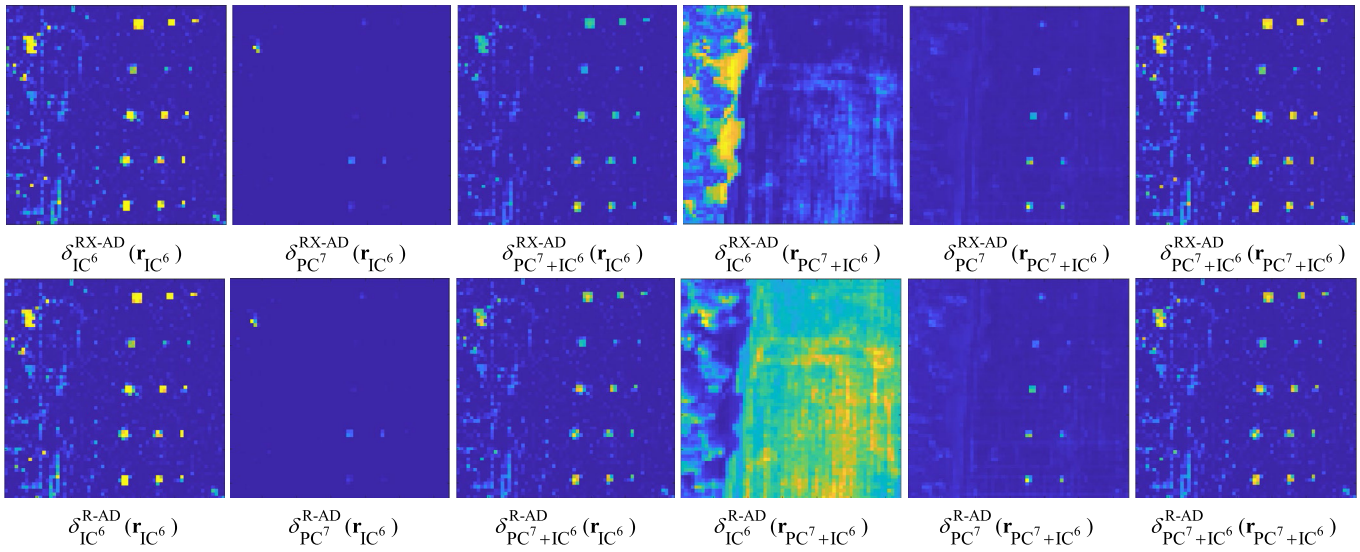


Fig. 8. Detection maps of the HYDICE 15-panel scene using CDASC with $p = 13$, $m = 7$, and $j = 6$.

1) HYDICR 15-Panel Scene: Figs. 7 and 8 show the detection maps of CDASC using the two sets of values of p, m, j, tabulated in Table I for the HYDICE 15-panel scene, respectively. Due to variable spatial sizes of panels in the HYDICE 15-panel scene, all the panel pixels in the first, second, and third columns are pure, mixed, and subpixels, respectively, where the best CDASC anomaly detectors were $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ by visual inspection. To conduct a quantitative analysis, Tables II and III tabulate the detection results in Figs. 5–7 using the eight 3-D ROC-curve-based detection measures, $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{ODP} derived in Section VI. Interestingly, the visual detection results did not reflect in the quantitative results in Tables II and III. For example, $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ had the best visual detection maps, but their corresponding AUC $_{(\text{D,F})}$ values were not the highest, which means that the commonly used criterion $\text{AUC}_{(\text{D,F})}$ does not really reflect the quality of the detection results rather than detector effectiveness in Tables II and III. On the other hand, $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$ and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ in Figs. 7 and 8 had the best BS and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ seemed to do better than $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$. However, the results in Tables II and III showed otherwise with $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$ being the best in $\text{AUC}_{(\text{F},\tau)}$, BS, and SNPR. Finally, for the over performance evaluation, $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ was the best in AUC_{TDBS} and AUC_{ODP}. Furthermore, the consistency of visual effects and quantitative evaluation verifies the results of the above detection criteria [3], [41], [45], [59].

The results in Tables II and III conveyed very important messages. Given the fact that 8 of 12 test anomaly detectors in Tables II and III produced $\text{AUC}_{(\text{D,F})}$ values approximately 0.99 within the error range of 10^{-3}, it is very difficult to assess which detector is best. Therefore, relying solely on $\text{AUC}_{(\text{D,F})}$ obtained from the traditional 2-D ROC curve of $(\text{P}_\text{D},\text{P}_\text{F})$ is misleading and unreliable.

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TABLE II
AUC OF HYDICE 15-PANEL SCENE USING CDASC WITH $p = 9$, $m = 5$, AND $j = 4$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9929	**0.8868**	0.0283	**1.8798**	0.9645	31.2730	**0.8585**	**1.8515**
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9903	**0.8868**	0.0284	1.8772	0.9618	31.1670	0.8584	1.8488
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9869	0.1383	**0.0009**	1.1252	**0.9860**	**151.0600**	0.1374	1.1243
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9845	0.1570	0.0012	1.1416	0.9833	133.8100	0.1559	1.1405
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9932	0.6659	0.0194	1.6592	0.9737	34.1620	0.6464	1.6397
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	**0.9934**	0.6090	0.0200	1.6025	0.9734	30.3770	0.5889	1.5825
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.6347	0.2056	0.0594	0.8403	0.5752	3.4565	0.1461	0.7808
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8478	0.3999	0.1885	1.2478	0.6592	2.1206	0.2113	1.0592
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8464	0.1944	0.0114	1.0409	0.8349	17.0220	0.1830	1.0294
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8230	0.2018	0.0150	1.0249	0.8080	13.4180	0.1867	1.0098
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9927	0.8456	0.0274	1.8384	0.9652	30.7940	0.8182	1.8109
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9932	0.6290	0.0221	1.6222	0.9710	28.3390	0.6068	1.6000

TABLE III
AUC OF HYDICE 15-PANEL SCENE USING CDASC WITH $p = 13$, $m = 7$, AND $j = 6$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9932	**0.8891**	0.0357	**1.8823**	0.9575	24.9010	**0.8534**	**1.8466**
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9932	**0.8891**	0.0357	**1.8823**	0.9574	24.8960	**0.8534**	**1.8466**
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9913	0.0613	**0.0010**	1.0527	**0.9902**	**55.9100**	0.0602	1.0516
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9911	0.0602	0.0011	1.0515	0.9900	55.7090	0.0592	1.0504
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9939	0.5536	0.0199	1.5475	0.9739	27.710	0.5336	1.5276
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9941	0.6564	0.0264	1.6506	0.9676	24.7930	0.6300	1.6242
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.5966	0.1311	0.1553	0.7277	0.4413	0.8445	-0.0241	0.5724
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8008	0.6682	0.4798	1.4691	0.3209	1.3926	0.1884	0.9892
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9707	0.2838	0.0147	1.2546	0.9560	19.2370	0.2691	1.2399
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9662	0.3228	0.0212	1.2891	0.9450	15.1780	0.3015	1.2679
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9922	0.8066	0.0351	1.7989	0.9571	22.9790	0.7715	1.7638
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	**0.9936**	0.7220	0.0312	1.6844	1.7157	0.9624	0.6907	1.6844

To further compare the recently developed anomaly detectors, CRD-DW-STO in [12] and OSPDS-AD in [41], Fig. 9 shows the detection maps of the best CDASC anomaly detectors in comparison to the best results by CRD-DW-STO with $(w_{\text{out}}, w_{\text{in}}) = (11, 7)$ in [12] and $\delta^{\text{OSPDS-AD}}_{\mathbf{L}_5}(\mathbf{r}_{\widehat{\mathbf{L}}_5+\widehat{\mathbf{S}}_4})$ in [41] where $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$ are the sphered space of the low-rank space $\mathbf{L}$

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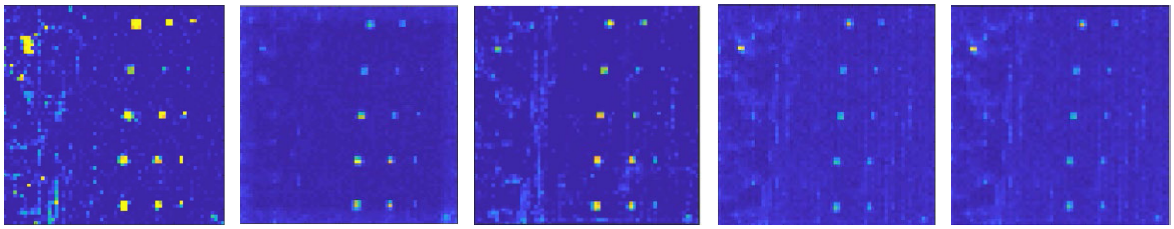


Fig. 9. Detection maps of CDASC, CRD-DW-STO, RX/R-AD, and OSPDS-AD for the HYDICE panel scene.

TABLE IV
COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, CRD-DW-STO, RX/R-AD, AND OSPDS-AD FOR HYDICE 15-PANEL DATA

Method	Detector	(p,m,j)	$\text{AUC}_{(\text{D,F})}$	$\text{AUC}_{(\text{D},\tau)}$	$\text{AUC}_{(\text{F},\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	(9, 5, 4)	0.9929	0.8868	0.0284	1.8798	0.9645	**31.2730**	**0.8585**	**1.8515**	1.5594
		(13, 7, 6)	0.9932	**0.8891**	0.0357	**1.8823**	0.9575	24.9010	0.8534	1.8466	2.0692
CRD	$(w_{\text{out}}, w_{\text{in}}) = (11,7)$	NA	**0.9953**	0.4673	0.0274	1.4626	**0.9679**	17.0547	0.4399	1.4352	4.7516
OSPDS-AD	$\delta^{\text{OSPDS-AD}}_{\text{L}_m}(\mathbf{r}_{\hat{\text{L}}_m+\hat{\text{S}}_j})$	(9, 5, 4)	0.9871	0.6398	**0.0243**	1.6269	0.9628	26.3314	0.6155	1.6026	78.6271
RX-AD	$\delta^{\text{RX-AD}}(\mathbf{r})$	NA	0.9898	0.3574	0.0430	1.3472	0.9468	8.3116	0.3144	1.3042	0.6187
R-AD	$\delta^{\text{R-AD}}(\mathbf{r})$	NA	0.9900	0.3584	0.0435	1.3484	0.9465	8.2391	0.3149	1.3049	0.6164

and the sparse space $\mathbf{S}$, respectively. By visual inspection of Fig. 9, the detection power of $\delta^{\text{RX-AD}}_{\text{IC}^4}(\mathbf{r}_{\text{IC}^4})$ and $\delta^{\text{RX-AD}}_{\text{S}_{\text{IC}^6}}(\mathbf{r}_{\text{S}_{\text{IC}^6}})$ was significantly better than those obtained by CRD-DW-STO with $(w_{\text{out}}, w_{\text{in}}) = (11, 7)$ and RX/R-AD. Table IV also shows their 3-D ROC-curve-derived eight detection measures where once again the overall performance of CDASC was better than all other test anomaly detectors.

Also tabulated in Table IV is the time complexity analysis in seconds where RX/R-AD required the least running time. Nevertheless, CDASC required much lesser time compared with CRD and OSPDS-AD.

2) HYDICE Urban Scene: Figs. 10 and 11 show the detection maps of the six anomaly detectors with the values of (p, m, j) according to Table I for the HYDICE urban scene. Except for few cases (such as $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ which seemed to produce the worst detection maps), it is very difficult to evaluate their detection performance by visual inspection.

Tables V and VI show the eight 3-D ROC-curve-derived detection measures for quantitative evaluation of 12 test anomaly detectors in terms of TD, BS, and detector's effectiveness. The results showed that $\delta^{\text{RX-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ and $\delta^{\text{RX-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ produced the best AUC values across the board for $(p, m, j) = (9, 5, 4)$ and $(p, m, j) = (13, 7, 6)$, respectively. These conclusions were different from that drawn for the HYDICE 15-panel scene. Nevertheless, $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$ which worked effectively for the HYDICE 15-panel scene were still among the best anomaly detectors for the HYDICE urban scene. What is more, $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ could suppress anomalies and detect BKG in the HYDICE urban scene as it did for the HYDICE 15-panel scene in Figs. 7 and 8. It seemed to be very obvious that the best anomaly detectors were $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ by visual inspection, which not only had better BS but also had better anomaly detectability. Similar to the HYDICE 15-panel scene, TD of $\delta^{\text{RX/R-AD}}_{\mathbf{IC}^j}(\mathbf{r}_{\mathbf{IC}^j})$ and $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m+\mathbf{IC}^j}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ was better than that of $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ but had more falsely alarm detected data samples in BKG. From the perspective of BS, $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{IC}^j})$ obtained the best results in the sense of $\text{AUC}_{(\text{F},\tau)}$ and AUC_{SNPR} but its anomaly detectability was poor. The quantitative results in Tables V and VI did confirm the visual results by accurately reflecting the ability of anomaly detectors in BS and TD.

Fig. 12 further shows the comparison of the detection maps of RX-AD using $\mathbf{A} = \mathbf{B} = \text{PC}^m + \text{IC}^j$ of CDASC to that obtained by CRD-DW-STO using $(w_{\text{out}}, w_{\text{in}}) = (11, 9)$ and $\delta^{\text{OSPDS-AD}}_{\text{L}_5}(\mathbf{r}_{\hat{\text{S}}_4})$. Table VII also tabulates their quantitative detection results by $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{ODP}. If we evaluate the test anomaly detectors solely based on $\text{AUC}_{(\text{D,F})}$, CRD was best with $\text{AUC}_{(\text{D,F})} = 0.9966$ followed by $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ with $\text{AUC}_{(\text{D,F})} = 0.9920$ within an error range of 0.005. However, other than $\text{AUC}_{(\text{D,F})}$, $\delta^{\text{OSPDS-AD}}_{\text{L}_5}(\mathbf{r}_{\hat{\text{S}}_4})$ produced the best values across board except $\text{AUC}_{(\text{F},\tau\text{t})} = 0.0248$ which was the fourth best values next to $\text{AUC}_{(\text{F},\tau\text{t})} = 0.0216$, $\text{AUC}_{(\text{F},\tau\text{t})} = 0.0232$, and $\text{AUC}_{(\text{F},\tau\text{t})} = 0.0233$ produced by $\delta^{\text{RX-AD}}_{\mathbf{PC}^5+\mathbf{IC}^4}(\mathbf{r}_{\mathbf{PC}^5+\mathbf{IC}^4})$, $\delta^{\text{RX-AD}}_{\mathbf{PC}^7}(\mathbf{r}_{\mathbf{PC}^7+\mathbf{IC}^6})$, and $\delta^{\text{RX-AD}}_{\text{S}_{\text{IC}^6}}(\mathbf{r}_{\text{S}_{\text{IC}^6}})$, respectively, and $\text{AUC}_{\text{BS}} = 0.9643$ which was the third best next to $\text{AUC}_{\text{BS}} = 0.9723$ and $\text{AUC}_{\text{BS}} = 0.9687$ produced by $\delta^{\text{RX-AD}}_{\text{S}_{\text{IC}^6}}(\mathbf{r}_{\text{S}_{\text{IC}^6}})$ and $\delta^{\text{RX-AD}}_{\mathbf{PC}^7}(\mathbf{r}_{\mathbf{PC}^7+\mathbf{IC}^6})$, respectively. Nevertheless, based on the overall performance evaluated by the eight detection measures, it is fair to say that $\delta^{\text{OSPDS-AD}}_{\text{L}_5}(\mathbf{r}_{\hat{\text{S}}_4})$ was the best anomaly detector despite the fact that CRD-DW-STO did produce the best value of $\text{AUC}_{(\text{D,F})}$ but unfortunately it produced the worst value of $\text{AUC}_{(\text{F},\tau)}$ and the second worst value of AUC_{BS}. Once again, these experiments further demonstrated that using $\text{AUC}_{(\text{D,F})}$ alone cannot provide real assessment of AD performance.

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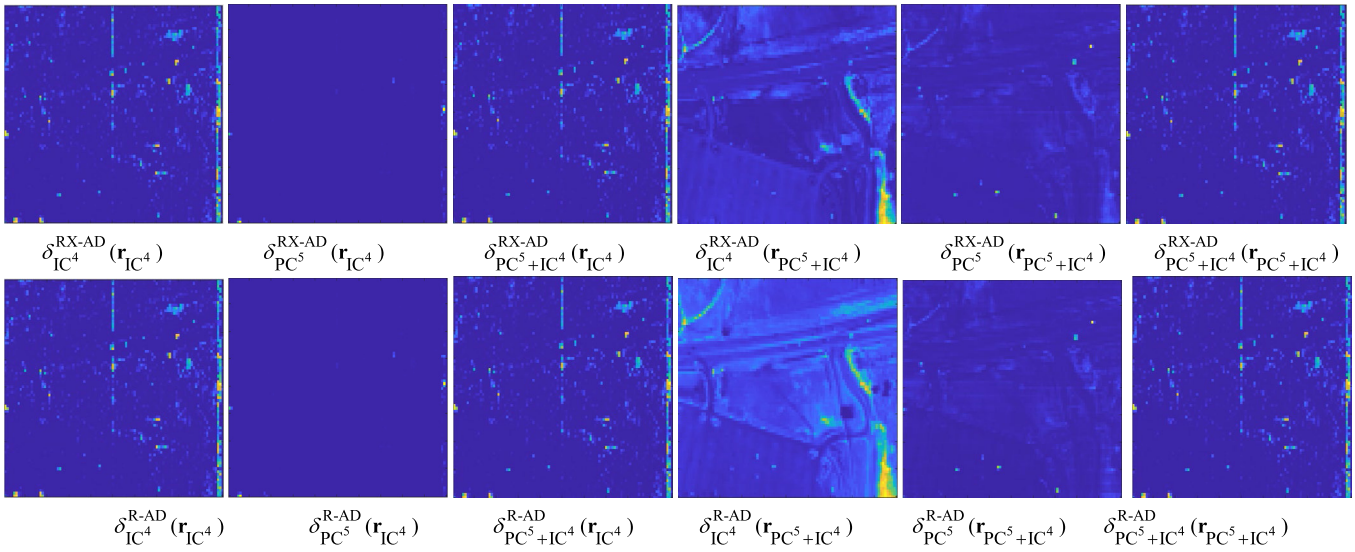


Fig. 10. Detection maps of the HYDICE urban scene using CDASC with $p = 9$, $m = 5$, and $j = 4$.

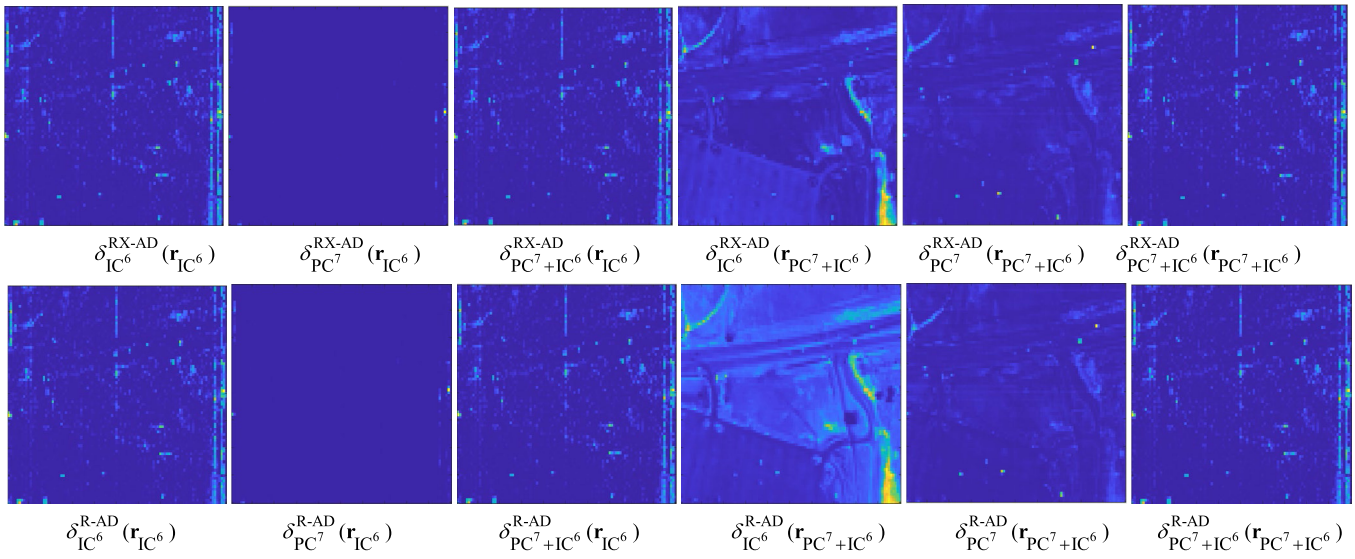


Fig. 11. Detection maps of the HYDICE urban scene using CDASC with $p = 13$, $m = 7$, and $j = 6$.

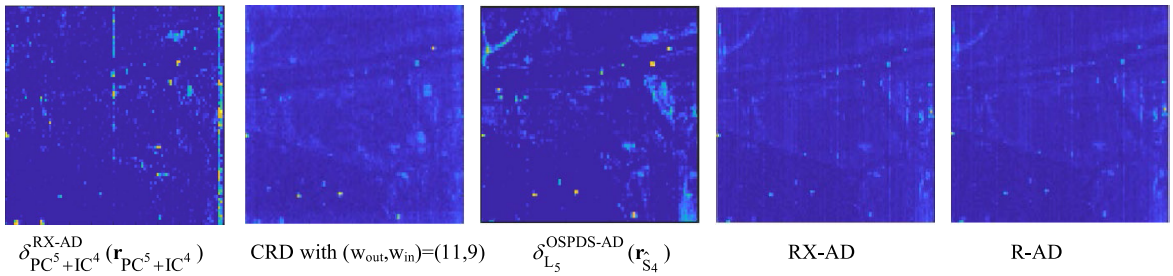


Fig. 12. Detection maps of CDASC, CRD-DW-STO, RX/R-AD, and OSPDS-AD for the HYDICE urban scene.

Similar to Table IV, the time complexity of RX/R-AD tabulated in Table VII was still the least followed by CDASC. The time complexity of OSPDS-AD was significantly increased due to the fact that the size of the HYDICE urban data was much larger than that of the HYDICE 15-panel data. Interestingly, CDASC and CRD did not dramatically increase the time complexity.

3) San Diego Airport Data: Technically speaking, it is hard to convince that the three airplanes in San Diego airport dataset in Fig. 6 are anomalous targets because they are huge

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TABLE V
AUC OF HYDICE URBAN DATA USING CDASC WITH $p = 9$, $m = 5$, AND $j = 4$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9438	0.3955	0.0212	1.3395	0.92262	18.5950	0.37431	1.3182
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9439	0.3956	0.02127	1.3395	0.9226	18.5920	0.3743	1.3183
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9310	0.0490	0.0008	0.9801	0.9302	60.0480	0.0482	0.9793
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9315	0.0490	**0.0008**	0.9805	0.9307	**60.1670**	0.0482	0.9797
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9439	0.3932	0.0211	1.3372	0.9227	18.5560	0.3720	1.3160
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9439	0.3929	0.0211	1.3369	0.9227	18.5410	0.3717	1.3157
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.7947	0.1350	0.0620	0.9297	0.7326	2.1766	0.0729	0.8676
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.7266	0.2350	0.1384	0.9617	0.5882	1.6982	0.0966	0.8233
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9287	0.3300	0.0194	1.2588	0.9092	16.9310	0.3105	1.2393
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9045	0.3254	0.0172	1.2300	0.8872	18.8940	0.3082	1.2128
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	**0.9586**	**0.3986**	0.0216	**1.3573**	**0.9369**	18.3730	**0.3769**	**1.3356**
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9584	0.3983	0.0216	1.3568	0.9368	18.4220	0.3767	1.3351

TABLE VI
AUC OF HYDICE URBAN DATA USING CDASC WITH $p = 13$, $m = 7$, AND $j = 6$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9726	0.3506	0.0232	1.3232	0.9493	15.0500	0.3273	1.2999
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9725	0.3506	0.0234	1.3232	0.9491	14.9600	0.3272	1.2998
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9345	0.0196	0.0009	0.9541	0.9336	20.2080	0.0186	0.9532
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.9371	0.0180	0.0009	0.9552	0.9361	18.0490	0.0170	0.9542
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9726	0.3487	0.0232	1.3215	0.9494	15.0160	0.3255	1.2982
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.9725	0.3415	0.0232	1.3140	0.9493	14.7200	0.3183	1.2908
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8210	0.1419	0.0563	0.9629	0.7647	2.5208	0.0856	0.9066
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.7198	0.2402	0.1433	0.9601	0.5765	1.6765	0.0969	0.8168
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	**0.9920**	**0.3714**	**0.0232**	**1.3635**	**0.9687**	**15.9650**	**0.3481**	**1.3402**
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9896	0.3526	0.0244	1.3422	0.9651	14.3970	0.3281	1.3178
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9758	0.3608	0.0243	1.3366	0.9515	14.8360	0.3364	1.3123
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.9751	0.3515	0.0240	1.3267	0.951	14.6150	0.3275	1.3027

visible targets. So, AD for this scene should be considered as unsupervised target detection. Nonetheless, since this dataset has been widely used for AD, it was also used for experiments. Figs. 13 and 14 show the detection maps of RX/R-AD using 12 CDA versions with (p, m, j) given in Table I where it is very obvious by visual inspection to see that the best

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TABLE VII

COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, CRD-DW-STO, RX/R-AD, AND OSPDS-AD FOR HYDICE URBAN SCENE

Method	Detector	(p,m,j)	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(9, 5, 4)	0.9586	0.3986	**0.0216**	1.3573	0.9369	**18.3730**	0.3769	1.3356	3.1347
	$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(13, 7, 6)	0.9920	0.3714	0.0232	1.3635	**0.9687**	15.9650	0.3481	1.3402	3.2660
CRD	$(w_{out},w_{in}) = (11, 9)$	NA	**0.9966**	**0.4469**	0.0506	**1.4435**	0.9460	8.8320	**0.3963**	**1.3928**	4.5929
OSPDS-AD	$\delta^{\text{OSPDS-AD}}_{\text{L}_m}(\mathbf{r}_{\text{S}_j})$	(9, 5, 4)	0.9892	**0.6377**	0.0248	**1.6268**	0.9643	**25.6640**	**0.6128**	**1.6020**	570.2245
RX-AD	$\delta^{\text{RX-AD}}(\mathbf{r})$	NA	0.9872	0.2575	0.0361	1.2447	0.9511	7.1330	0.2214	1.2087	1.7303
R-AD	$\delta^{\text{R-AD}}(\mathbf{r})$	NA	0.9872	0.2549	0.0359	1.2421	0.9513	7.1003	0.2190	1.2062	**1.6745**

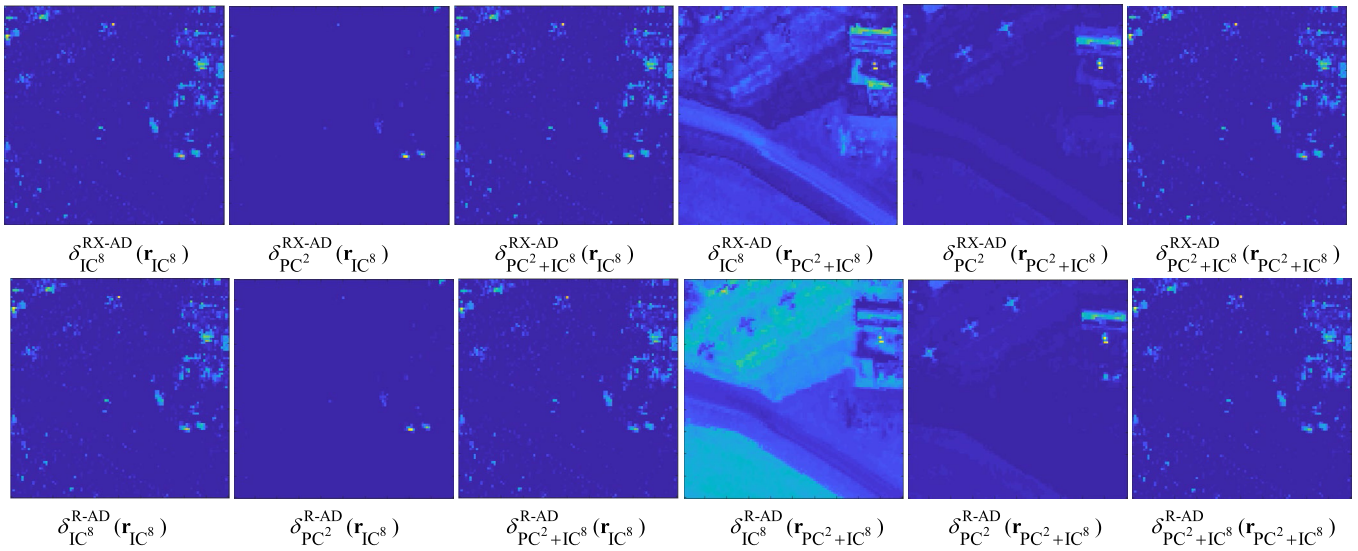


Fig. 13. Detection maps of the San Diego scene using CDASC with $p = 10$, $m = 2$, and $j = 8$.

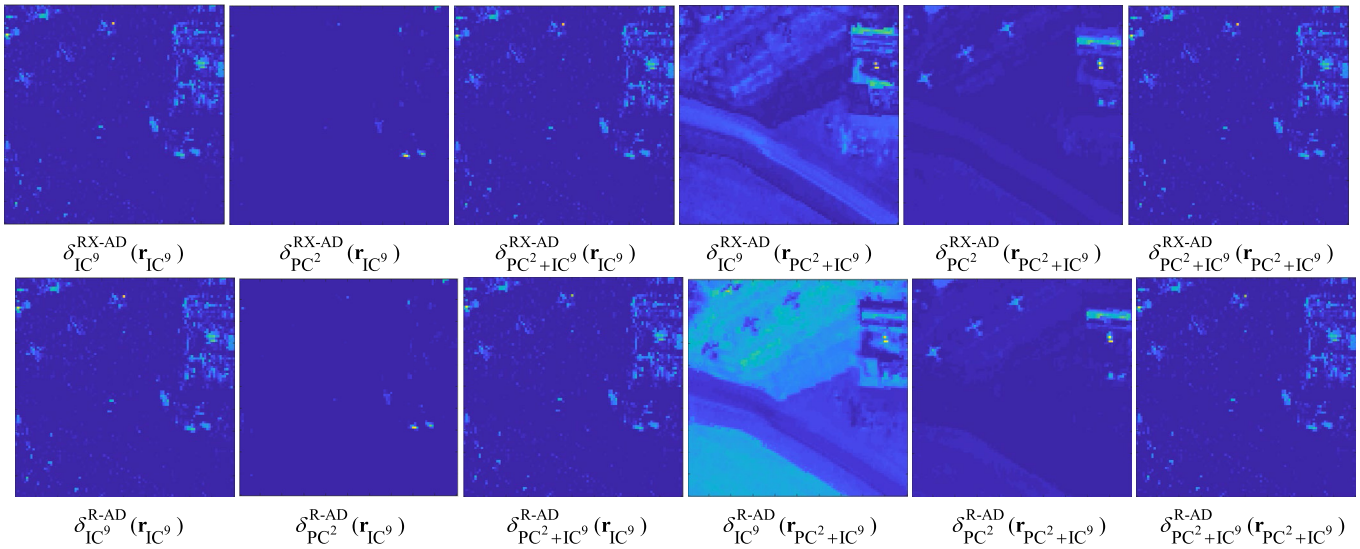


Fig. 14. Detection maps of the San Diego scene using CDASC with $p = 11$, $m = 2$, and $j = 9$.

anomaly detectors were $\delta^{\text{RX/R-AD}}_{\mathbf{PC}^m}(\mathbf{r}_{\mathbf{PC}^m+\mathbf{IC}^j})$ which suppressed BKG by $\mathbf{PC}^m$ and detected targets in $\mathbf{PC}^m+\mathbf{IC}^j$. As expected, the conclusions drawn from this scene were different from that concluded from the HYDICE 15-panel scene and HYDICE urban scene.

Tables VIII and IX show their quantitative detection results of Figs. 13 and 14 by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{ODP} where $\delta^{\text{RX-AD}}_{\text{IC}^8}(\mathbf{r}_{\text{IC}^8})$ and $\delta^{\text{RX-AD}}_{\text{PC}^2+\text{IC}^8}(\mathbf{r}_{\text{PC}^2+\text{IC}^8})$, which performed well in both the HYDICE datasets in Sections VIII-A and VIII-B,

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TABLE VIII
AUC OF SAN DIEGO DATA USING CDASC WITH $p = 10$, $m = 2$, AND $j = 8$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8262	0.0885	0.0192	0.9147	0.807	4.6093	0.0693	0.8954
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8257	0.0885	0.0192	0.9142	0.8065	4.6093	0.0693	0.8950
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.7291	0.0041	0.0019	0.7332	0.7272	2.1578	0.0022	0.7313
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.7213	0.0040	0.0019	0.7253	0.7194	2.1052	0.0021	0.7234
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8264	0.0882	0.0191	0.9146	0.8073	4.6178	0.0691	0.8956
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8258	0.0834	0.0184	0.9092	0.8074	4.5326	0.065	0.8908
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.5389	0.0871	8.79E-02	0.626	0.451	0.9908	-0.0008	0.5382
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.3656	0.1822	0.2636	0.5478	0.102	0.6911	-0.0814	0.2842
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	**0.9010**	**0.1596**	**2.09E-02**	**1.0606**	**0.8801**	**7.6363**	**0.1387**	**1.0398**
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8902	0.1536	0.0209	1.0438	0.8693	7.3492	0.1327	1.0229
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8373	0.0895	0.0192	0.9268	0.8181	4.6614	0.0703	0.9075
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8361	0.0846	0.0185	0.9207	0.8176	4.5729	0.0661	0.9022

TABLE IX
AUC OF SAN DIEGO DATA USING CDASC WITH $p = 11$, $m = 2$, AND $j = 9$

Detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8493	0.0866	0.0195	0.9359	0.8298	4.441	0.0671	0.9164
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8490	0.0866	0.0195	0.9356	0.8295	4.441	0.0671	0.9161
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.7542	0.0046	0.0023	0.7588	0.7519	2	0.0023	0.7565
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{IC}^j})$	0.7435	0.0045	0.0023	0.748	0.7412	1.9565	0.0022	0.7458
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8494	0.0860	0.0193	0.9354	0.8301	4.456	0.0667	0.9161
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{IC}^j})$	0.8493	0.0848	0.0192	0.9341	0.8301	4.4167	0.0656	0.9150
$\delta^{\text{RX-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.4735	0.0701	8.09E-02	0.5436	0.3926	0.8665	-0.0108	0.4628
$\delta^{\text{R-AD}}_{\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.3825	0.1808	0.2480	0.5633	0.1345	0.729	-0.0672	0.3153
$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	**0.9010**	**0.1596**	**0.0209**	**1.0606**	**0.8801**	**7.6364**	**0.1387**	**1.0398**
$\delta^{\text{R-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8902	0.1536	0.0209	1.0438	0.8693	7.3493	0.1327	1.0229
$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8570	0.0873	0.0195	0.9443	0.8375	4.4769	0.0678	0.9248
$\delta^{\text{R-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	0.8564	0.0860	0.0193	0.9424	0.8371	4.456	0.0667	0.9231

did not work well for this airport scene as we anticipated. This is mainly because the large airplanes can be captured by PCs rather than ICs. This phenomenon further validated an interesting finding that the San Diego data are not suitable for AD, but more suitable for target detection.

Once again, Fig. 15 shows the detection maps of RX/R-AD using $\mathbf{A} = \text{PC}^m + \text{IC}^j, \mathbf{B} = \text{PC}^m$ in comparison to

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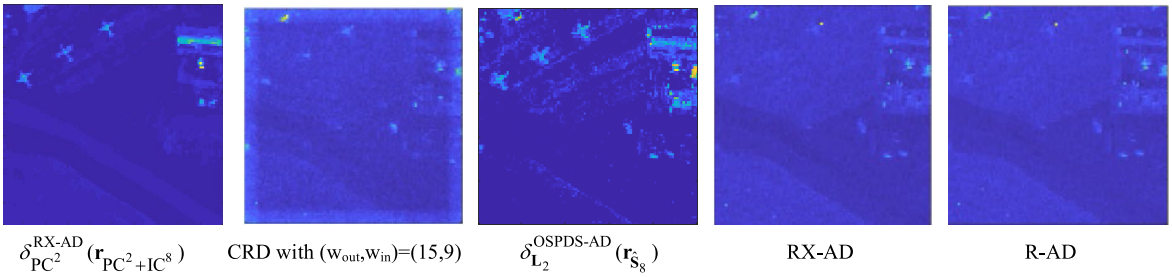


Fig. 15. Detection maps of CDASC, CRD-DW-STO, RX/R-AD, and OSPDS-AD for the San Diego data scene.

TABLE X
COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, CRD-DW-STO, RX/R-AD, AND OSPDS-AD FOR SAN DIEGO DATA

Method	Detector	(p,m,j)	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(10, 2, 8)	**0.9010**	**0.1596**	**0.0209**	**1.0606**	**0.8801**	7.6363	**0.1387**	**1.0398**	4.1880
		(11, 2, 9)	**0.9010**	**0.1596**	**0.0209**	**1.0606**	**0.8801**	**7.6364**	**0.1387**	**1.0398**	4.3514
CRD	$(w_{out},w_{in}) = (15, 9)$	NA	0.8942	0.1546	0.0589	1.0488	0.8353	2.6248	0.0957	0.9900	30.2724
OSPDS-AD	$\delta^{\text{OSPDS-AD}}_{\mathbf{L}_m}(\mathbf{r}_{\hat{\mathbf{S}}_j})$	(10, 2, 8)	**0.9463**	**0.2495**	0.0260	**1.1958**	**0.9202**	**9.5863**	**0.2235**	**1.1698**	1104.166
RX-AD	$\delta^{\text{RX-AD}}(\mathbf{r})$	NA	0.8314	0.0802	0.0453	0.9116	0.7861	1.7704	0.0349	0.8663	**1.9622**
R-AD	$\delta^{\text{R-AD}}(\mathbf{r})$	NA	0.8270	0.0832	0.0501	0.9102	0.7769	1.6607	0.0331	0.8602	1.9690

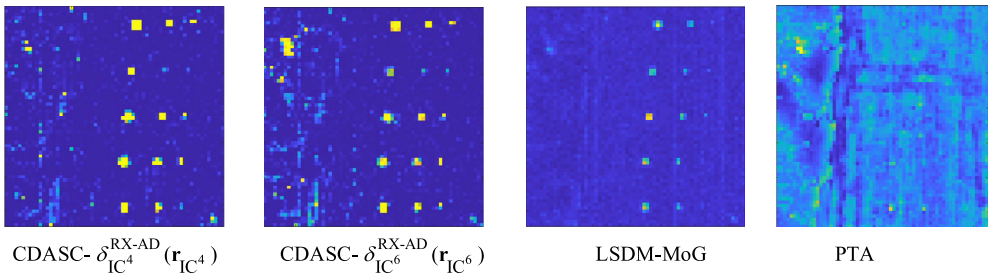


Fig. 16. AD maps produced by CDASC, LSDM-MoG, and PTA for the HYDICE panel data.

CRD-DW-STO and OSPDS-AD along with their corresponding eight quantitative results tabulated in Table XI where the best performance came from $\delta^{\text{OSPDS-AD}}_{\mathbf{L}_2}(\mathbf{r}_{\hat{\mathbf{S}}_8})$ followed by CDASC. In particular, CDASC performed better than CRD for all the eight detection measures across board. These experiments also demonstrated that the San Diego airport data scene is not suitable for AD.

According to Table X, the time complexity of OSPDS-AD was significantly higher than other test anomaly detectors due to the large size of the San Diego airport scene. CRD also required high time complexity because it was directly related to the data size and its use of window size. In addition, CDASC was also affected by the image size to some extent, but their time complexities were still maintained relatively low.

Finally, it should be noted that CRD-DW-STO [12] and OSPDS-AD in [41] have shown to perform better than many other anomaly detectors. In this case, only CRD-DW-STO and OSPDS-AD were compared to CDASC-AD. We refer more details to [41].

VIII. COMPARATIVE STUDIES AND ANALYSES

As reviewed in Section I, extensive studies have been devoted to AD in the literature. Many proposed works have developed very complicated models and systems with a large number of fine-tuned parameters and their values without guidelines but rather determined by trial and error empirically. As a result, it is very difficult to repeat their experiments.

In this section, two most recent articles [35] and [37] were selected to conduct comparative studies and analyses because their works pertain to CDASC-AD and their algorithms are available on website that can be repeated. Specifically, some datasets used in these works were indeed the same as the datasets used in this article. In this case, we can perform creditable comparisons for performance evaluation based on our proposed CDASC-AD results against their published results, which are believed to be the best results obtained by fine-tuning their used parameters.

In the LRaSMD model (3), the low-rank and sparse spaces are decomposed by GoDec using SC to constrain anomalies in its found sparse space $\mathbf{S}$. While [35] still used a low-rank and sparse decomposition model (LSDM) similar to the LRaSMD model, it used an MoG noises in the sparse component so as to derive LSDM-MoG-MD using the MD to perform AD. Since a key to success in hyperspectral AD is how to effectively distinguish anomalies from the background, especially when BKG is complex and anomalies are weak. For this reason,

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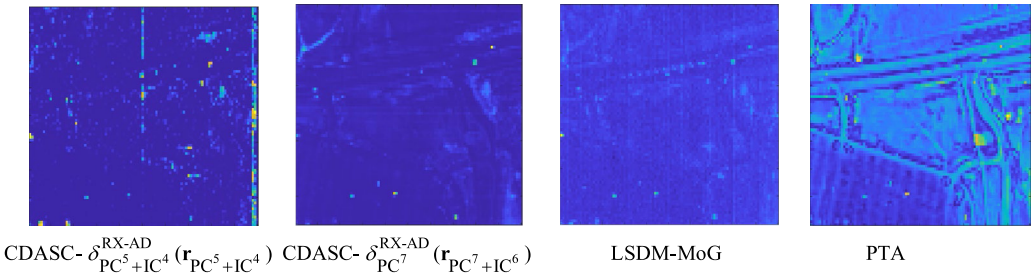


Fig. 17. AD maps produced by CDASC, LSDM-MoG, and PTA for the HYDICE urban data.

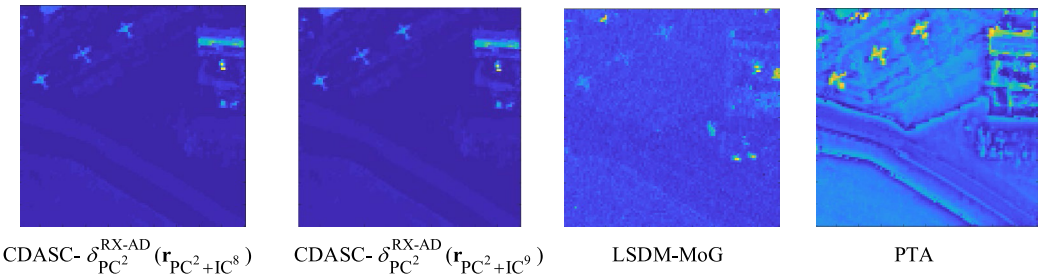


Fig. 18. AD maps produced by CDASC, LSDM-MoG, and PTA for the San Diego data.

the PTA developed in [37] was selected to perform anther comparative analysis where PTA decomposed a hyperspectral imagery into a background tensor and an anomaly tensor corresponding to BKG and anomaly spaces, respectively.

Figs. 16–18 show the detection results obtained by LSDM-MoG-MD and PTA for the HYDICE 15-panel scene, HYDICE urban, and San Diego airport datasets, respectively, where the best results of CDASC-AD obtained in Section VII-B are also included for comparison. By visual inspection, it is very clear that LSDM-MoG performed better than PTA for HYDICE panel and urban dataset, while PTA detected three airplanes more visibly than LSDM-MoG but with many falsely alarmed data samples. Nevertheless, overall speaking, CADSC-AD did outperform both LSDM-MoG and PTA, particularly, significantly better than PTA.

To conduct a detailed quantitative analysis among CDASC, LSDM-MoG, and PTA, the 3-D ROC analysis developed in Section VI was used as an evaluation tool for comparative studies. Using the AD maps in Figs. 16–18, we generated 3-D ROC curves along with their corresponding three 2-D ROC curves, ROC curves of (P_D,P_F), ROC curves of (P_D,τ), and ROC curves (P_F,τ) for the three datasets as shown in Figs. 19–21. It should be noted that since the two versions of CDASC-AD, CDASC-$\delta^{\text{RX-AD}}_{\text{PC}^8}(\mathbf{r}_{\text{PC}^2+\text{IC}^8})$ and CDASC-$\delta^{\text{RX-AD}}_{\text{PC}^2}(\mathbf{r}_{\text{PC}^2+\text{IC}^9})$, both produced identical results for the San Diego data, only three ROC curves were generated. For the HYDICE data, PTA performed worst by all counts, while LSDM-MoG did best in BS. As TD, CDASC did best as shown by the ROC curves of (P_D,τ). On the whole, CDASC performed better than LSDM-MoG and PTA. Similar conclusions could be drawn for the HYDICE urban data. Interestingly, the experiments for the San Diego data showed otherwise where PTA did best in the ROC curves of (P_D,P_F) and ROC curves of (P_D,τ) but worst in the ROC curves of (P_F,τ). This experiment explained why PTA was reported to be best in its ROC curve of (P_D,P_F) and concluded that PTA performed well in AD in [37]. Unfortunately, it did not report its worst performance in its ROC curve of (P_F,τ).

To further conduct quantitative analysis based on the ROC curves in Figs. 18–21, Tables XI–XIII also tabulate the quantitative detection results calculated by $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, $\text{AUC}_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{ODP} for the three datasets. Also shown in Tables XI–XIII, the best results are boldfaced, and the results by LSDM-MoG-MD and PTA are highlighted by yellow for better visual comparison. Since the used HYDICE 15-panel data of [35, Fig. 6] were exactly the same as the one in Fig. 4 of this article with the scene being flipped over by 90° clockwise rotation. A detailed analysis can be conducted. According to the results in Tables XI–XIII, LSDM-MoG-MD did better than CDASC-AD in terms of $\text{AUC}_{(D,F)}$ for the HYDICE 15-panel data but worse than CDASC-AD in all other seven detection measures. So, if we evaluate AD performance solely based on the value of $\text{AUC}_{(D,F)}$, it is easy to jump to conclusion that LSDM-MoG-MD performed better than CDASC-AD. However, as shown in [35, Fig. 16(c)], two subpixel panels at the bottom row were missed by detection and many falsely alarmed pixels were caused by poor BS. Unfortunately, these phenomena were not reported. This was because $\text{AUC}_{(D,F)}$ could not effectively evaluate the AD performance in BS. Without appealing for the 3-D ROC analysis, Li *et al.* [35] could only evaluate $\text{AUC}_{(D,F)} = 0.99630$ in Table I and $P_F = 0.0214$ with $P_D = 1$ in Table II and $P_D = 0.847$ with $P_F = 0.01$ in Table III, and also P_F with a specified $P_D = 1$ in Table II and P_D with a specified $P_F = 0.01$ in Table III. Now, if we use all eight detection measures in (25)–(29) to quantify TD, BS, and overall performance as tabulated in Tables XI–XIII, we immediately find that

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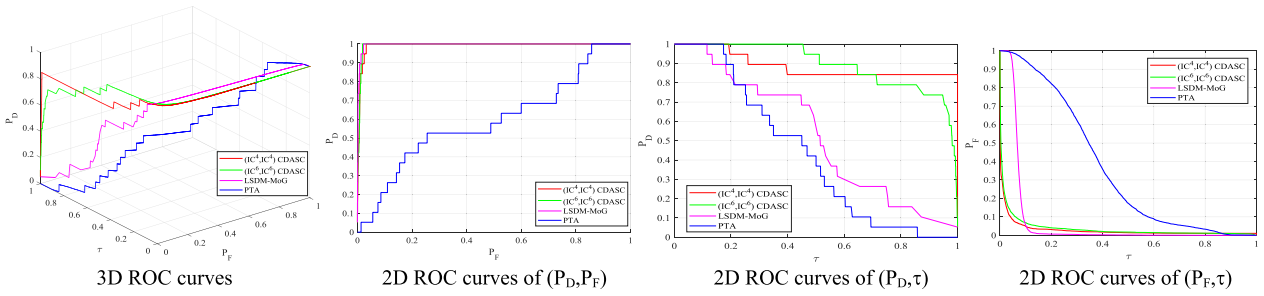


Fig. 19. Three-dimensional ROC curves along with the corresponding 2-D ROC by CDASC, LSDM-MoG-MD, and PTA for the HYDICE 15-panel data.

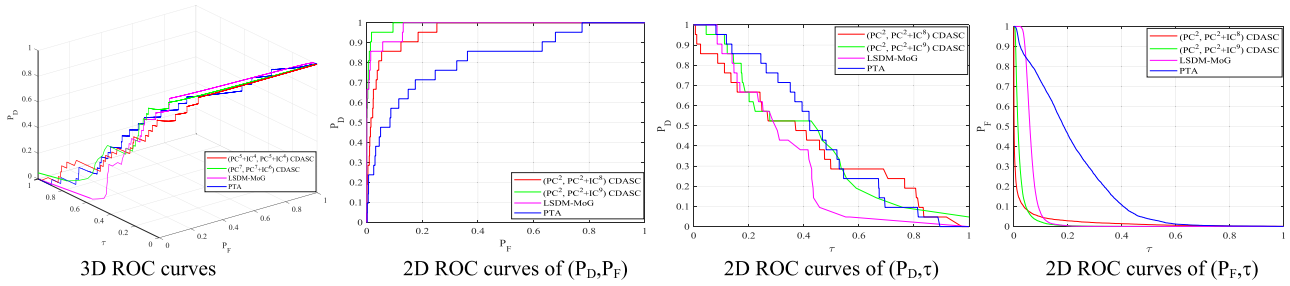


Fig. 20. Three-dimensional ROC curves along with the corresponding 2-D ROC by CDASC, LSDM-MoG-MD, and PTA for the HYDICE urban scene.

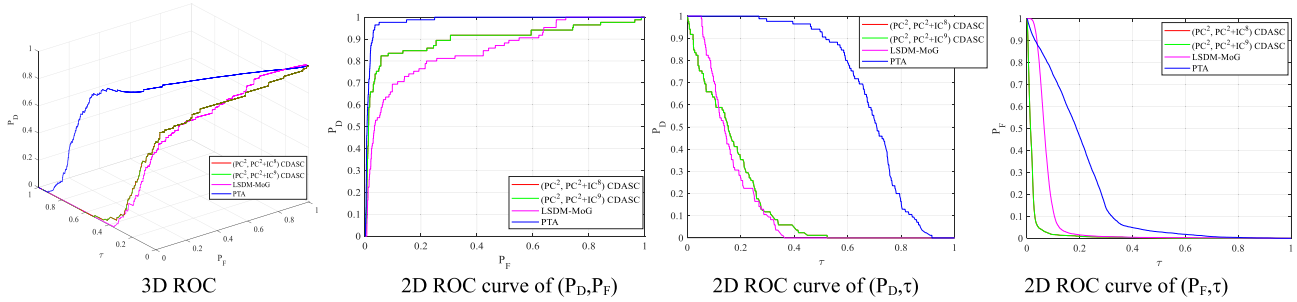


Fig. 21. Three-dimensional ROC curves along with the corresponding 2-D ROC by CDASC, LSDM-MoG-MD, and PTA for the San Diego data.

TABLE XI
COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, PTA, AND LSDM-MOG FOR HYDICE 15-PANEL DATA

Method	Detector	(p,m,j)	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta_{IC^j}^{RX\text{-}AD}(\mathbf{r}_{IC^j})$	(9, 5, 4)	0.9929	0.8868	**0.0284**	1.8798	**0.9645**	**31.2730**	**0.8585**	**1.8515**	**1.55**
		(13, 7, 6)	0.9932	**0.8891**	0.0357	**1.8823**	0.9575	24.9010	0.8534	1.8466	2.06
LSDM-MoG	$Rank_0$=20	NA	**0.9962**	0.5241	0.0694	1.5203	0.9268	7.5484	0.4547	1.4508	5.30
PTA	α=1.7, β=0.069	NA	0.5953	0.3739	0.2799	0.9692	0.3155	1.3360	0.0940	0.6894	11.60

CDASC-AD in fact performed BS very well, while LSDM-MoG-MD did not. For the other two datasets, CDASC-AD also outperformed LSDM-MoG-MD all the way across board in Tables XI–XIII.

There are two reasons attributed to why CDASC-AD performed better than LSDM-MoG-MD. One is that CDASC-AD uses the CDASC model to generate ICs as a sparse component space **S** by SC which can take care of non-Gaussian noise, generally assumed to be the case in hyperspectral imagery. The other is that RX/R-AD took advantage of separate low-rank and sparse spaces via (23) and (24) to suppress BKG and extract anomalies from the SC-constrained sparse space, $\mathbf{IC}_{CDASC}^{j}$.

As for PTA performance, according to Tables XI–XIII, PTA did best for overall performance for the San Diego airport data among all test methods but worst in BS. Unfortunately, PTA performed much worse for the other two datasets compared with CDASC-AD and LSDM-MoG in terms of every detection measure. Generally speaking, from the eight calculated detection measures, PTA performed very poorly across board, specifically BS. The major reason that PTA did not work effectively as expected is simply because it did not separate noise as a third noise tensor due to the fact noise does not have spatial information and a prior cannot be incorporated into the noise tensor. This is the same issue encountered in model (2) by RPCA in [31]. In contrast, CDASC-AD took

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TABLE XII
COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, PTA, AND LSDM-MoG FOR HYDICE URBAN SCENE

Method	Detector	(p,m,j)	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta^{\text{RX-AD}}_{\text{PC}^m+\text{IC}^j}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(9, 5, 4)	0.9586	0.3986	**0.0216**	1.3573	0.9369	**18.3730**	0.3769	1.3356	**3.13**
	$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(13, 7, 6)	**0.9920**	0.3714	0.0232	1.3635	**0.9687**	15.9650	0.3481	1.3402	3.26
LSDM-MoG	Rank_0=20	NA	0.9748	0.3178	0.0680	1.2926	0.9068	4.6719	0.2498	1.2246	12.62
PTA	α=1.7, β=0.069	NA	0.7990	**0.4457**	0.2265	1.2447	0.5725	1.9676	0.2192	1.0182	20.50

TABLE XIII
COMPARATIVE PERFORMANCE ANALYSIS BETWEEN CDASC, PTA, AND LSDM-MoG FOR SAN DIEGO DATA

Method	Detector	(p,m,j)	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time (s)
CDASC	$\delta^{\text{RX-AD}}_{\text{PC}^m}(\mathbf{r}_{\text{PC}^m+\text{IC}^j})$	(10, 2, 8)	0.9010	0.1596	**0.0209**	1.0606	**0.8801**	**7.6363**	0.1387	1.0398	**4.18**
		(11, 2, 9)	0.9010	0.1596	**0.0209**	1.0606	**0.8801**	**7.6364**	0.1387	1.0398	4.35
LSDM-MoG	Rank_0=20	NA	0.8559	0.2685	0.2030	1.1245	0.6530	1.3229	0.0655	0.9215	9.25
PTA	α=1.7, β=0.069	NA	**0.9734**	**0.6892**	0.2358	**1.6626**	0.7376	2.9224	**0.4533**	**1.4268**	28.79

care of noise by imposing SC on the ICA-generated space. In addition, PTA did well for the San Diego airport dataset because the airplanes are large and strong targets in which case noise does very little on PTA.

It seems that most recent works on AD used $AUC_{(D,F)}$ as the only criterion to evaluate their AD performance and then claimed their methods better than others by producing higher $AUC_{(D,F)}$ values such as [35] and [37]. As a matter of fact, many of them which produced very high values of $AUC_{(D,F)}$ had indeed very low values of $AUC_{(F,\tau)}$, AUC_{BS}, and AUC_{SNPR}. Concluding their superiority solely based on $AUC_{(D,F)}$ is misleading. More such examples can be found in [3], [41], [45], [59], and [61]. In particular, in [59] the adaptive cosine estimate (ACE), which has been considered as one of the commonly known best target detectors, had very high $AUC_{(D,F)}$ values for the HYDICE 15-panel data and San Diego airport, but actually poorest BS in Figs. 2(j) and 4(h).

A final comment is worthwhile. According to the above comparative analysis, the 3-D ROC analysis developed in [45] is in fact a very powerful and useful evaluation tool for AD. Since AD does not have any prior knowledge, BS is a very important performance indicator to evaluate its performance. The experiments performed in this article showed that many anomaly detectors which produce very high $AUC_{(D,F)}$ may actually turn out to have very poor BS as illustrated by the above two examples, LSMD-MoG-MD and PTA.

IX. CONCLUSION

This article presents a new approach to AD, called CDA-based AD, as an alternative to many current approaches, specifically LRaSR-based methods. CDA integrates two popular component analysis models, PCA and ICA, to derive a new model, called CDA model specified by $\mathbf{PC}^m + \mathbf{IC}^j + \mathbf{N}$ where $\mathbf{PC}^m$ and $\mathbf{IC}^j$ are the orthogonal subspace complements with m and j being the number of PCs and number of ICs, respectively. In particular, $\mathbf{PC}^m$ are components that can be used to characterize BKG and $\mathbf{IC}^j$ are components that can retain signal sources as anomalies. To separate anomalies from noise to enhance AD, the concept of SC used by GoDec is further incorporated into CDA to derive a CDASC model which was shown by experiments to outperform the existing anomaly detectors. For CDA to be better used, the key issue is how to determine appropriate values of m and j. Instead of directly determining these two values, VD is first used to estimate the value of $p = m + j$ and then is followed up by MX-SVD to determine the value of j which finally gives the value of m. Interestingly, both VD and MX-SVD are also eigen-based analysis techniques. As a result, the developed CDA theory can be considered as an eigen-analysis-based theory. With these estimated values of m and j, CDASC-AD works very effectively and CDASC can be a better model than the LRaSR-based models.

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